

AI in Genomics 2026

Tunisia · 14–18 Sept 2026

Introduction to structural modelling with AlphaFold

Presenter: Andrés Posbeyikian

Andrés Posbeyikian, Yu Sugihara, Amal Boukteb,
Jame Tannous, James Canham, Sophien Kamoun

PROGRAMME



01 **Predict**



02 Interpret



03 Compare



04 Design

What is genomics—the science of DNA

- ➡ The genome is the book of life
- ➡ How useful is a book if it's bound?
You need to open it and read it.
- ➡ Genomics and related technologies enable *reading*, *editing* and *re-writing* the book



What is genomics—the science of DNA

- ➡ *reading*: extracting information from the genome sequence
- ➡ *editing*: making targeted changes to the genome
- ➡ *re-writing*: constructing synthetic genes and genomes



From Discovery to Design...

- ➡ ...from discovery science (*reading*) to engineering biology (*editing* and *re-writing*).
- ➡ The ultimate promise of genomics is to shift biology from a descriptive to a design-driven discipline, where life becomes a programmable material.

Today's focus is reading the genome

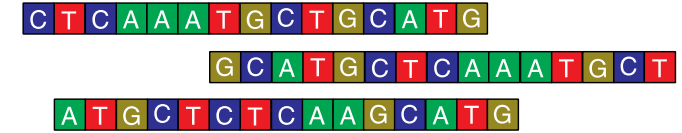
- ➡ *reading*: extracting information from the genome sequence
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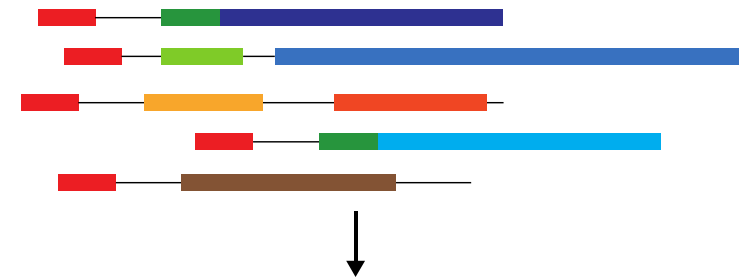
What is reading the genome?

- ❖ Gene prediction
- ❖ Variant calling (SNPs, indels)
- ❖ Comparative genomics
- ❖ Regulatory element mapping
- ❖ Transcriptome analysis
- ❖ Repeats
- ❖ Phylogenetic analysis
- ❖ Functional annotation
- ❖ Protein motifs

(a) Sequencing of the plant pathogen genome



(b) Computational prediction of pathogen effectors



(c) Library of effector clones



(d) *In planta* expression of effectors

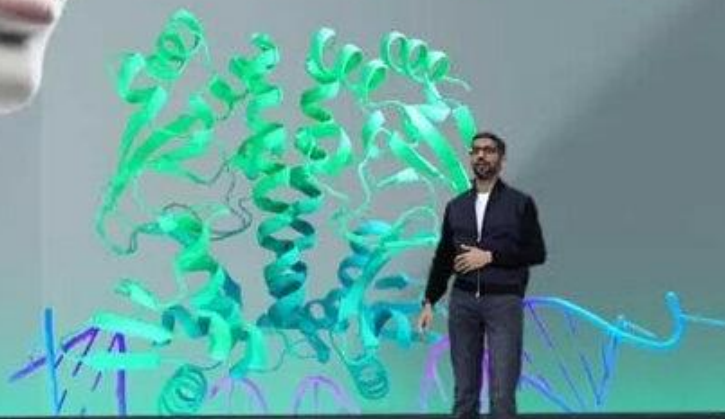




Google DeepMind

INTRODUCING

AlphaFold-3



The Nobel Prize in Chemistry 2024

They cracked the code for proteins' amazing structures

The Nobel Prize in Chemistry 2024 is about proteins, life's ingenious chemical tools. David Baker has succeeded with the almost impossible feat of building entirely new kinds of proteins. Demis Hassabis and John Jumper have developed an AI model to solve a 50-year-old problem: predicting proteins' complex structures. These discoveries hold enormous potential.

Related articles

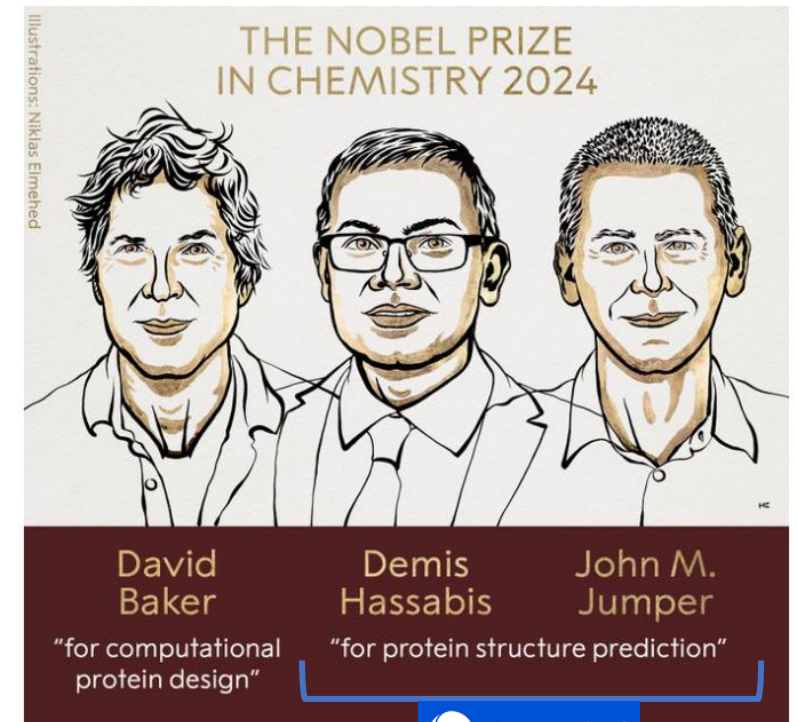
[Press release](#)

[Popular information: They have revealed proteins' secrets through computing and artificial intelligence](#)

[Scientific background: Computational protein design and protein structure prediction](#)



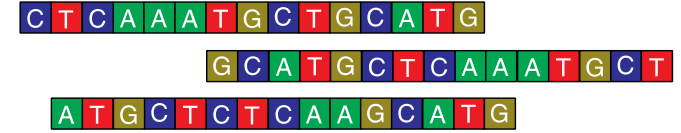
© Johan Jarnestad/The Royal Swedish Academy of Sciences



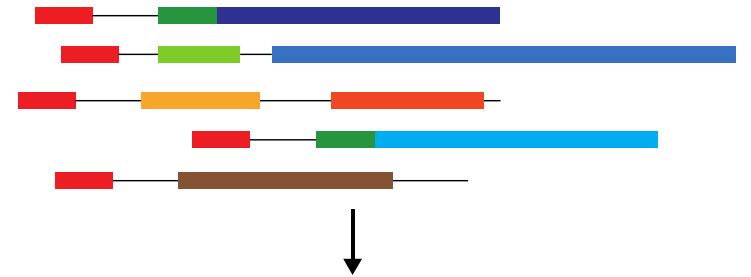
What is reading the genome?

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- ❖ Protein motifs
- ❖ Protein structure modelling
- ❖ Protein-Protein Interactions

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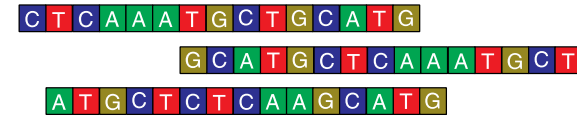
(d) *In planta* expression of effectors



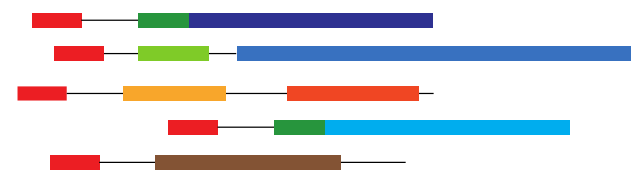
You can get even more out of your genome sequences

- ❖ Gene prediction
- ❖ Variant calling (SNPs, indels)
- ❖ Comparative genomics
- ❖ Regulatory element mapping
- ❖ Transcriptome analysis
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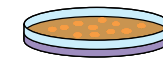
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2. Interpreting confidence metrics

3. Strengths and Limitations (AF2 and AF3)

4. Tips & tricks

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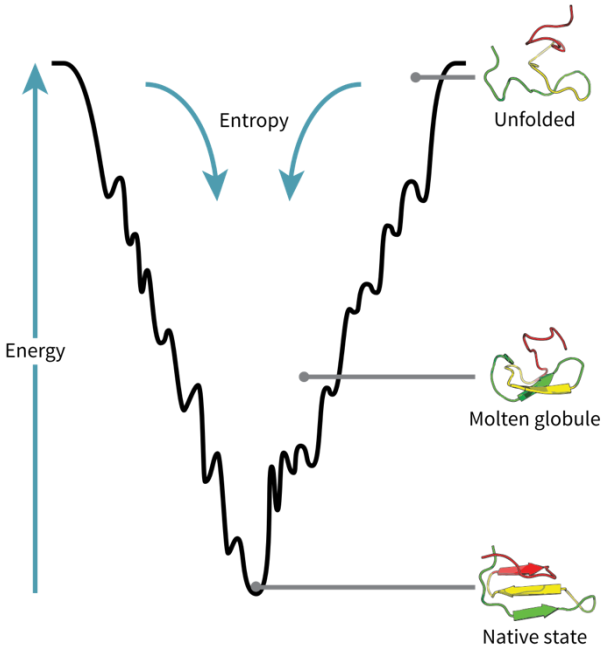
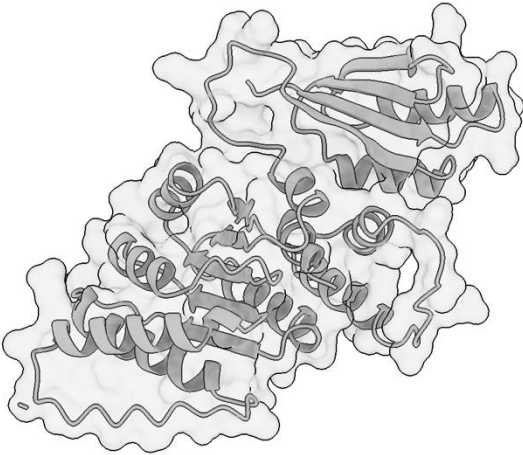
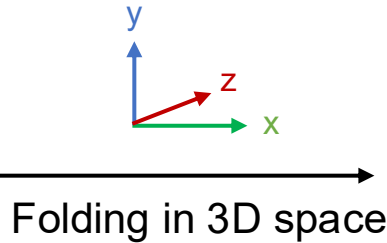
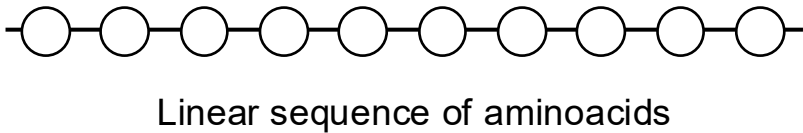
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The problem of protein folding...



Folding funnel hypothesis, Wikipedia.

Critical Assessment of Structure Prediction (CASP)

“The Olympics” of protein folding...

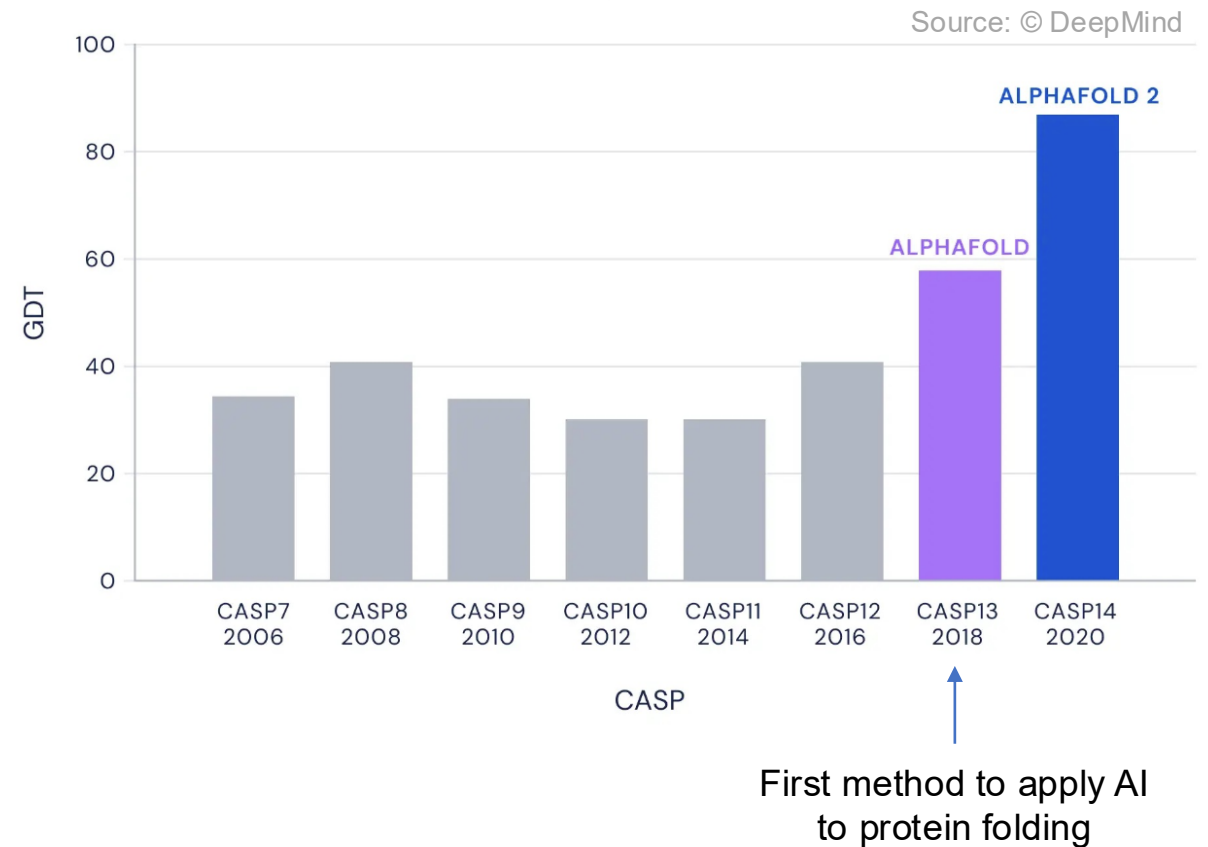
- Community-wide experiment for protein structure prediction
- It has been held every two years since 1994.
- **Objective:** to test structure prediction methods and **evaluate the state of the art** in protein structure modeling.
- Provides independent evaluation to the research community and software users.
- Viewed by many as a "**world championship**" in the field of protein structure prediction.
- More than 100 research groups from around the world participate regularly.
- Entire groups suspend their research to focus on preparing their servers and making detailed predictions.

Critical Assessment of Structure Prediction (CASP)

“The Olympics” of protein folding...

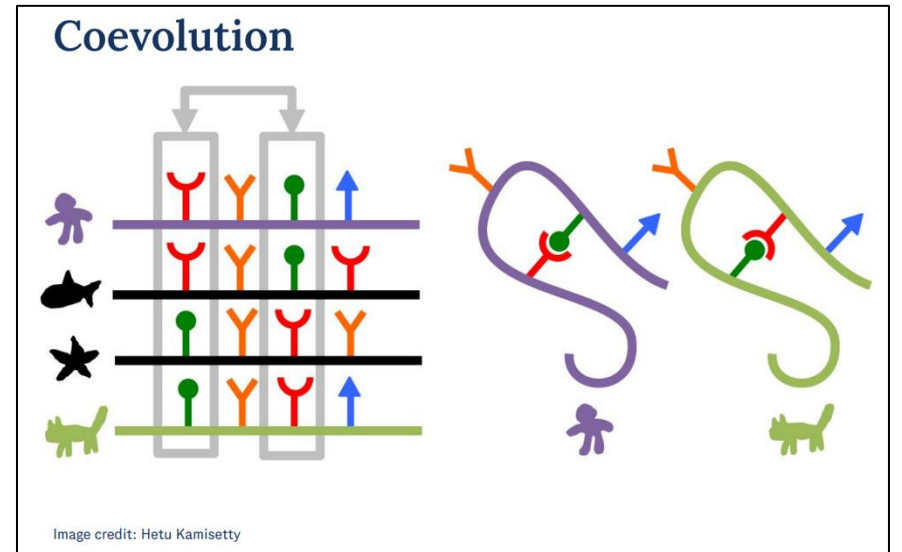
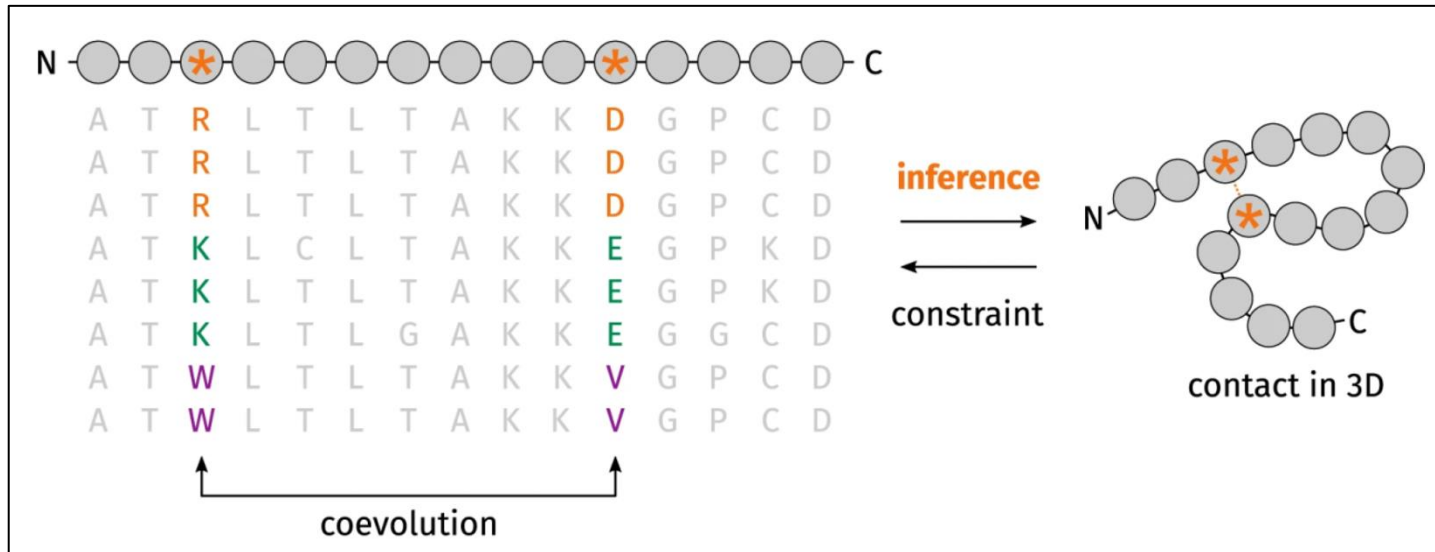
- **Dec-2018:** AlphaFold (AF), created by DeepMind, made headlines by winning CASP13.
- **Nov-2020:** An improved version, AlphaFold2, won CASP14.
- AF2 scored around 90 points on a 100-point prediction accuracy scale for moderately difficult protein targets.
- AlphaFold2 became **open source** in 2021.
- CASP15 in **2022:** Although DeepMind did not participate, virtually all highly ranked teams used AlphaFold or modifications of AlphaFold.

Median Free-Modelling Accuracy



AlphaFold. What is it and what are the principles behind it?

Co-evolutionary analyses inform structure



Schematic of how co-evolution methods extract information about protein structure from a multiple sequence alignment (MSA).

Adapted from **Oxford Protein Informatics Group**,
AlphaFold 2 is here: what's behind the structure prediction miracle

Can three-dimensional contacts in protein structures be predicted by analysis of correlated mutations?

I.N.Shindyalov¹, N.A.Kolchanov¹ and C.Sander²

¹Institute of Cytology and Genetics, Russian Academy of Sciences, Siberian Department, Prospect Lavrentyeva 10, Novosibirsk 630090, Russia and

²EMBL, Meyerhofstrasse 1, D-69012 Heidelberg, Germany

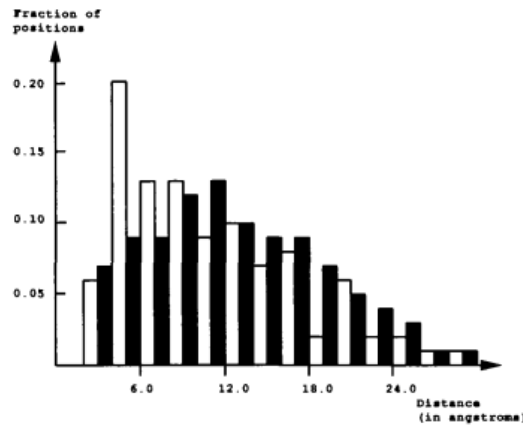
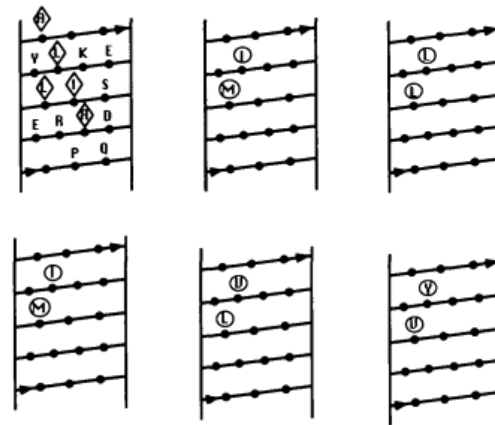


Fig. 3. Evidence for weak information about spatial closeness in correlated residue pairs. Histogram of the fraction of residue pairs as a function of spatial distance. The subset of mutationally correlated pairs (white) tends to occur at smaller spatial distances than do all pairs (black). Mutational correlation for particular residue pairs is assessed using equation (12), with a cut-off of p_0 of 0.20. Residue distance is the distance of closest interatomic approach between side chains. The numbers are accumulated for the eight proteins in Table III.



96	P	Q	E	R	A	D	L	S	V	L	K	E	A	109
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AlphaFold2. What is it, and how does it work?

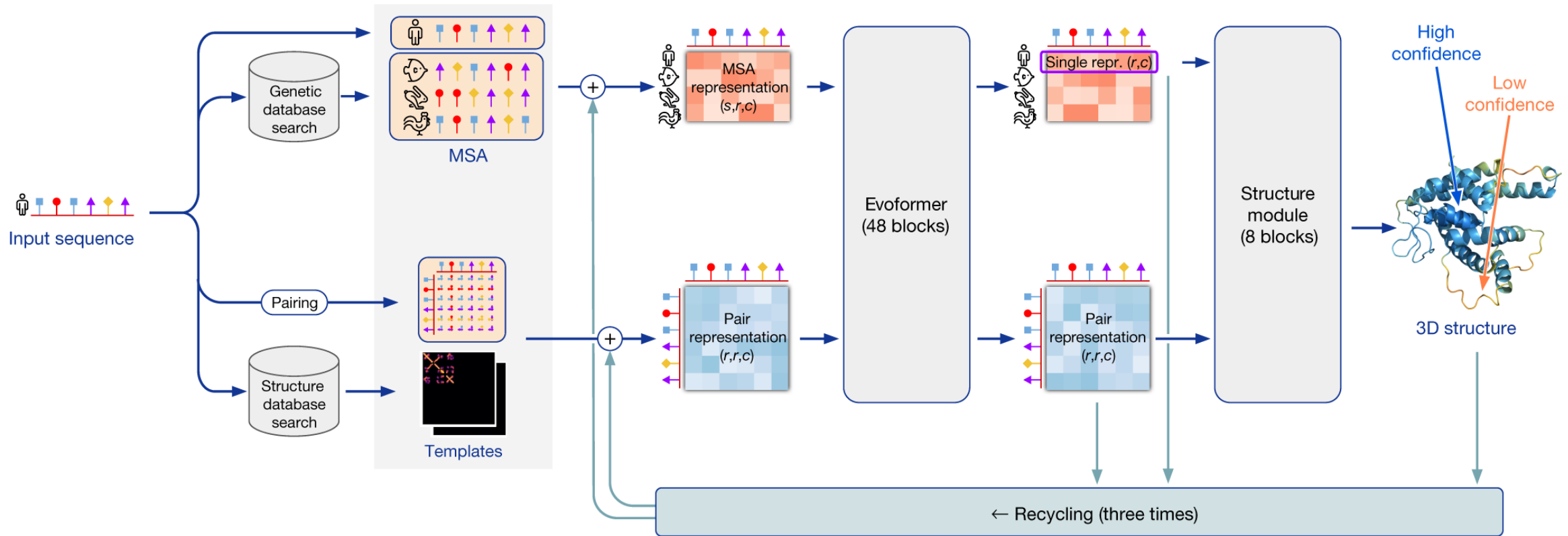


Diagram of AlphaFold 2 as officially published in Jumper *et. al* .2021, PMID: 34265844

AlphaFold2. What is it, and how does it work?

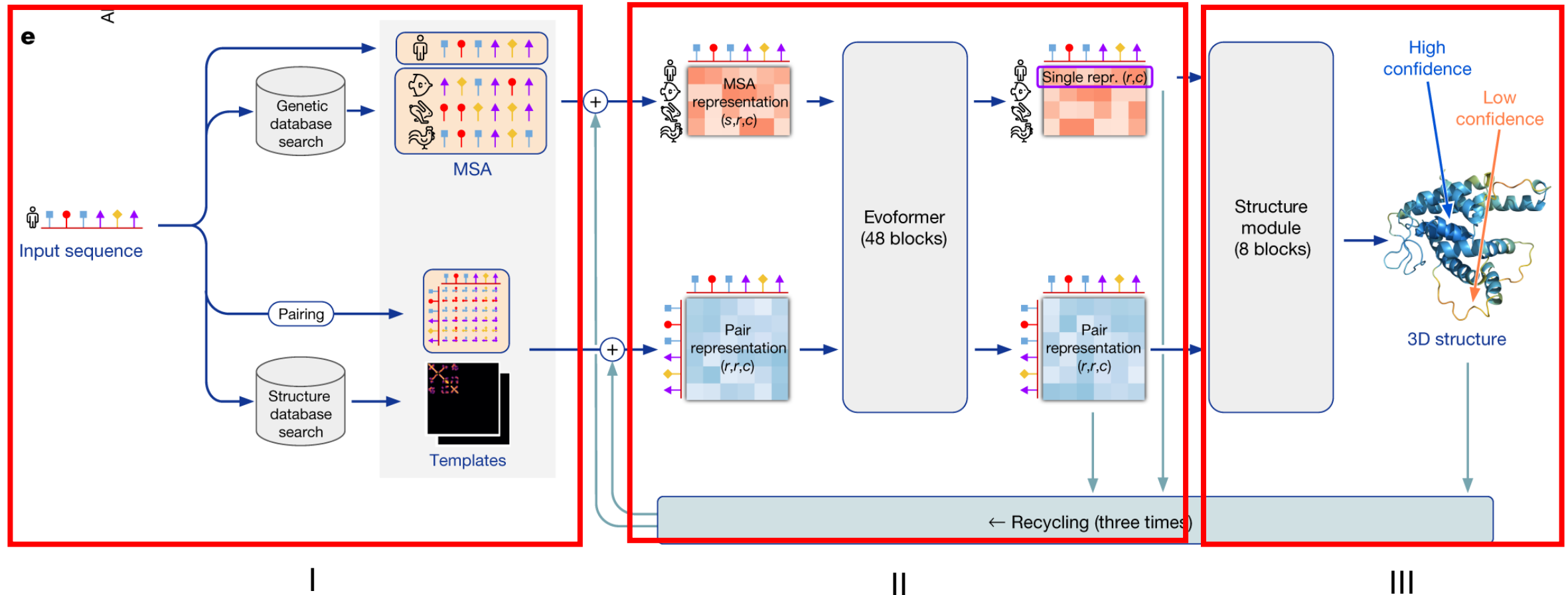
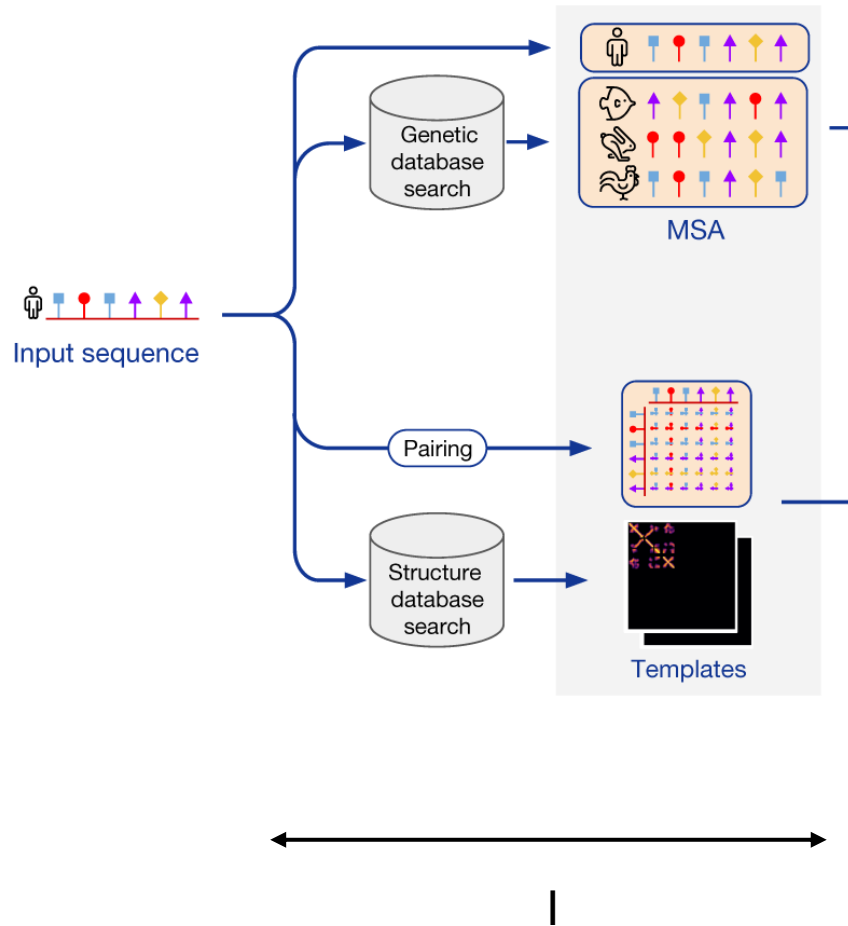


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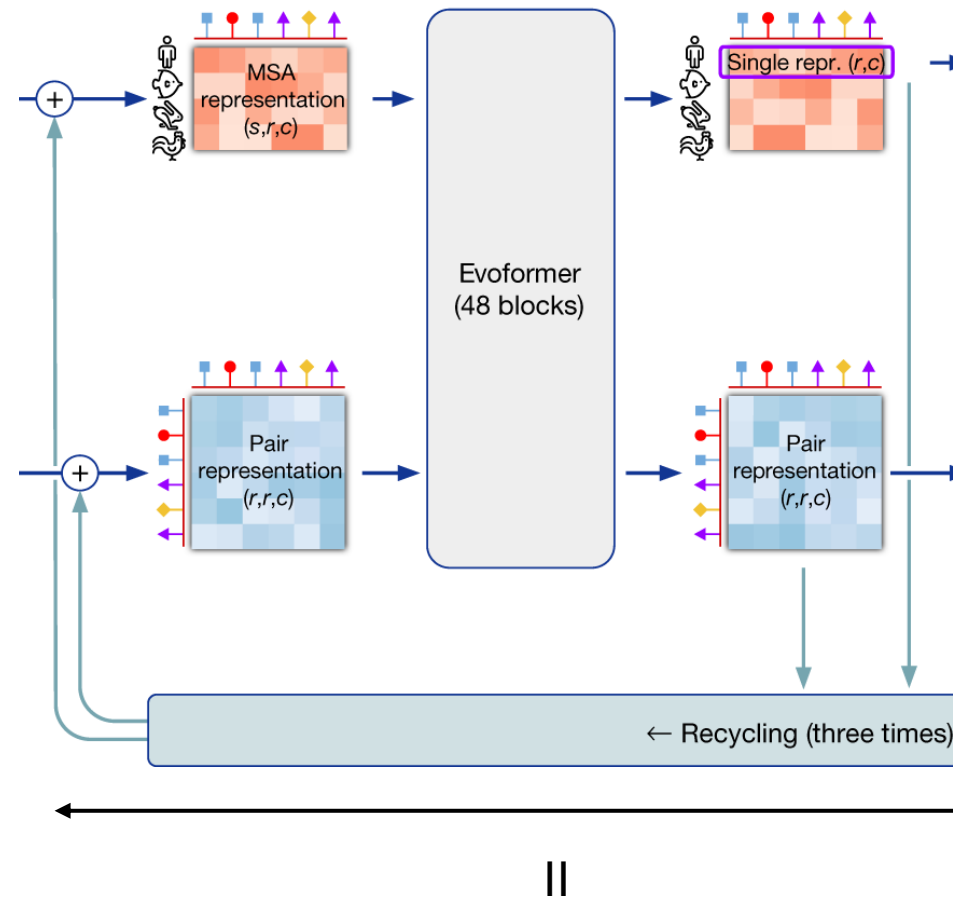
AlphaFold2. What is it, and how does it work?



- AF2 takes the input sequence of a.a. and searches sequence databases, building a multiple sequence alignment (**MSA**).
- The **MSA** identifies similar but non-identical sequences, found across organisms.
- This allows it to determine **parts of the sequence that are more likely to mutate, and to detect correlations between them**.
- AF2 also builds an initial 3D representation of the structure, called "**pairwise distance plot**", where it saves the predicted distances between pairs of a.a. This representation is essentially a **model of which aminoacids are likely to be in contact**.
- AF2 can be used as is (template-free), or we can suggest proteins that may have a similar structure to our input ("**templates**"), to bias the modelling. *This is also a feature present in AF3, which we will talk about later.*

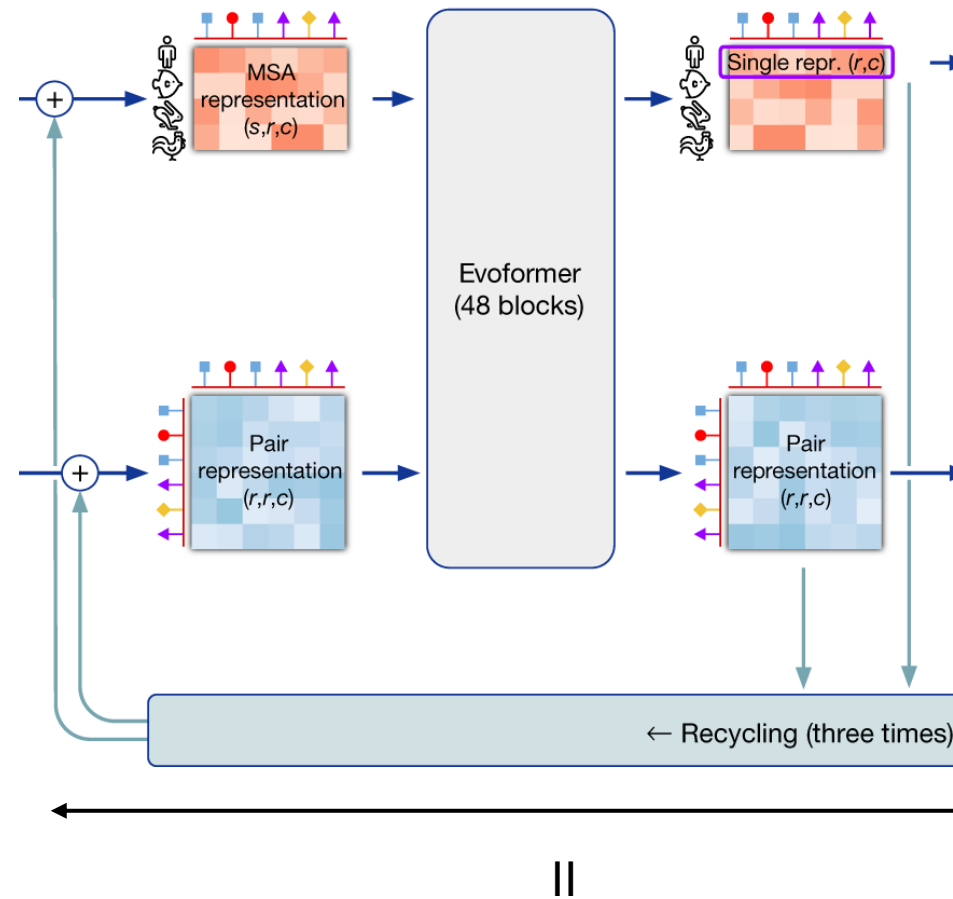
AlphaFold2. What is it, and how does it work?

- Once the MSA y de pairwise distance plot matrices are built, they enter a neuronal network called **Evoformer**.
- This block identifies what pieces of data are most informative.
- The **objective** of this module is to refine the MSA and pairwise representations, but also to **exchange information iteratively between them**.



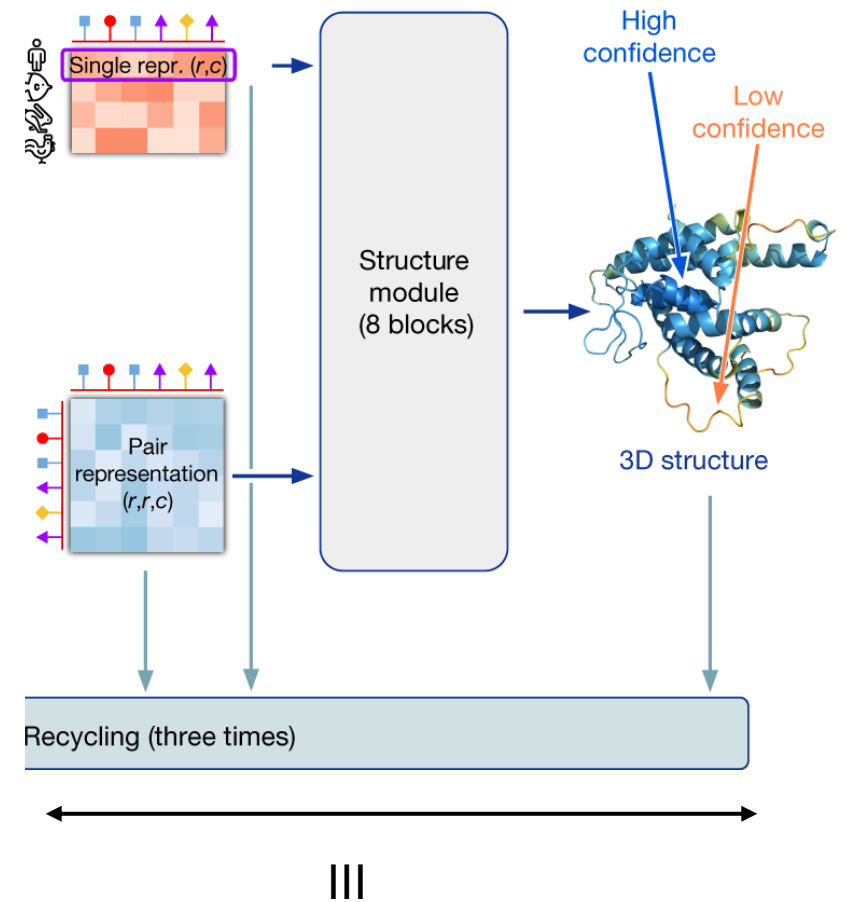
AlphaFold2. What is it, and how does it work?

- A better MSA improves the characterization of the geometry of the model, which simultaneously helps refine the MSA.
- This process is organized in blocks that repeat iteratively through 48 cycles.

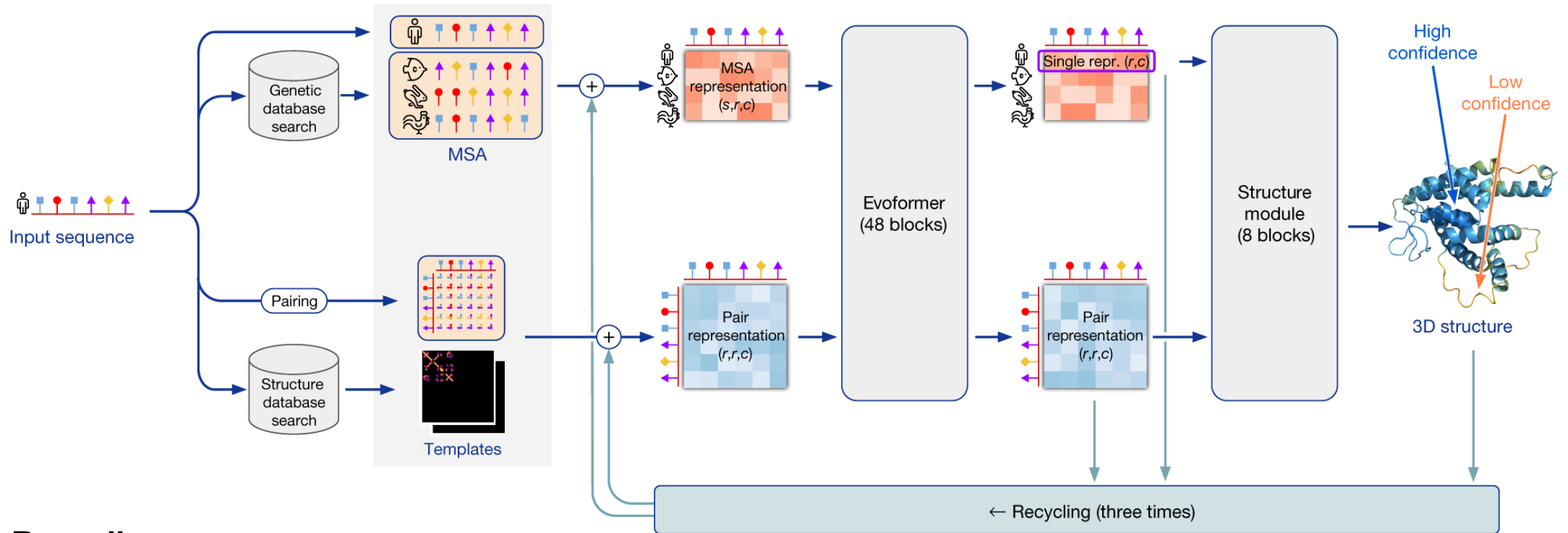


AlphaFold2. What is it, and how does it work?

- Finally, the MSA and pairwise distance matrices enter a second neural network called "**Structure module**", which generates a 3D model of the protein.
- This algorithm doesn't iterate but instead generates a single model in one step.
- The result is a list of coordinates with the position of each atom of the protein, including side chains.



AlphaFold2. What is it, and how does it work?



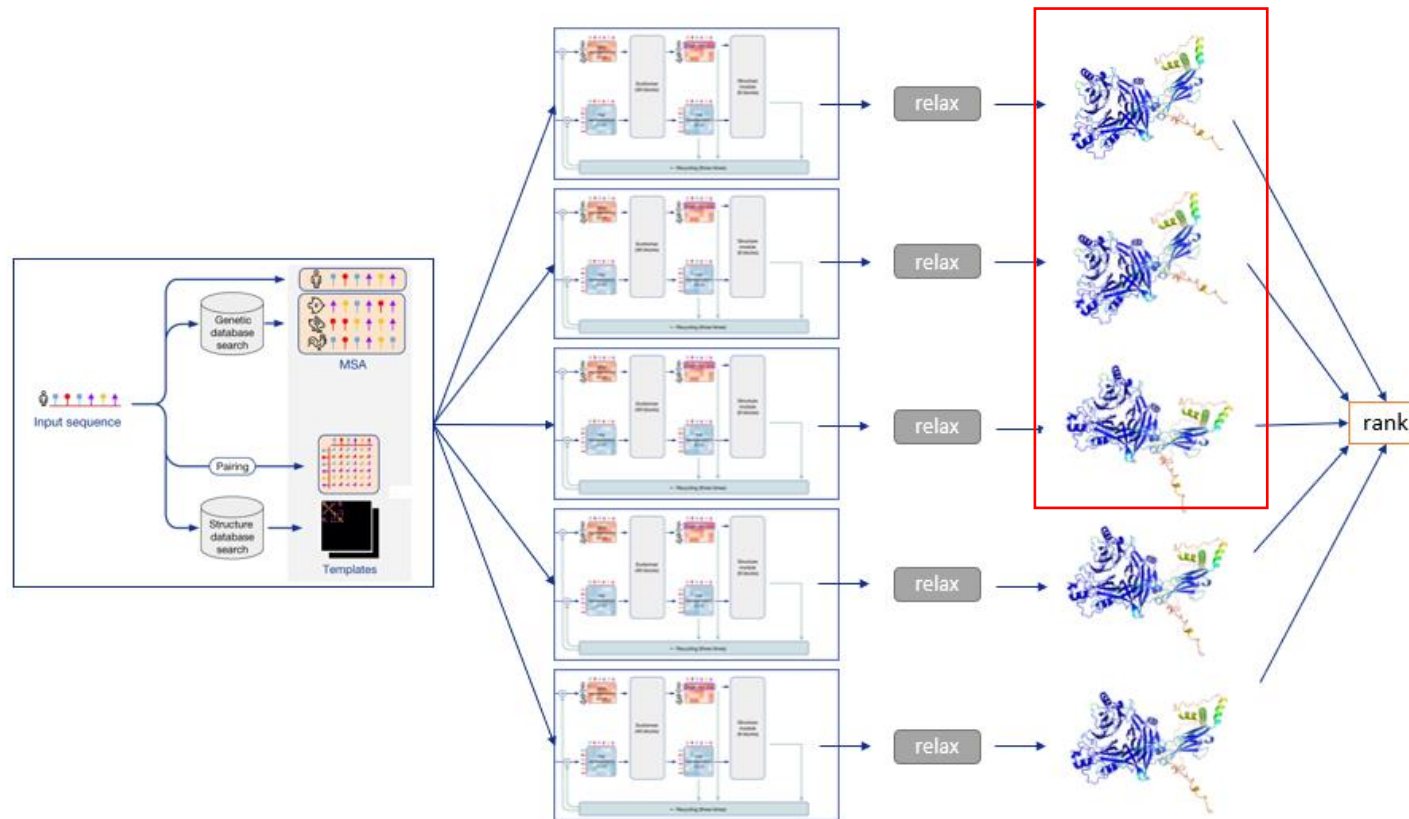
Recycling:

The algorithm works iteratively: after generating a final structure, it takes all the information (i.e., MSA, pair representation, and predicted structure) and feeds it back through the Evoformer block, recycling three times.

This allows the model to **refine its predictions**. *Benchmarking studies have shown that recycling can be increased up to 12 times, which significantly improves the reliability of the predicted structures.*

AlphaFold2. What is it, and how does it work?

The algorithm trains five different models using different random seeds, some with **templates** and some without, to encourage **diversity in the predictions**.



It assigns a ranking to the five models based on a metric called average **pLDDT**, which we will look at later. **In general**, people tend to pay more attention to the top 3 models generated by AF2.

AlphaFold Protein Structure Database

Developed by DeepMind and EMBL-EBI

Search for protein, gene, UniProt accession or organism

BETA

Search

Examples: [Free fatty acid receptor 2](#) [At1g58602](#) [Q5VSL9](#) [E. coli](#) Help: [AlphaFold DB search help](#)

Feedback on structure: [Contact DeepMind](#)

Contents

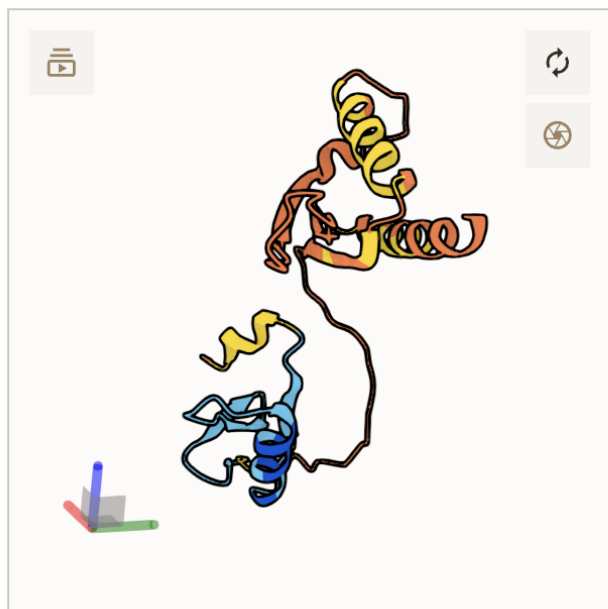
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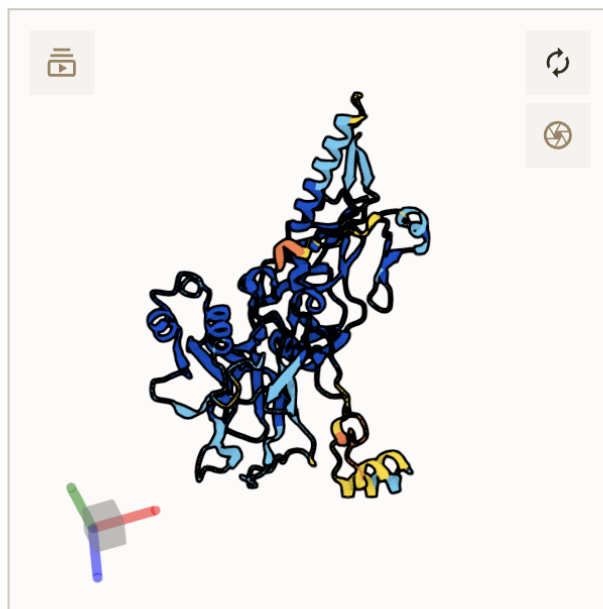
4. Tips & tricks

Not all predictions have the same degree of confidence and it's important to be able to tell them apart



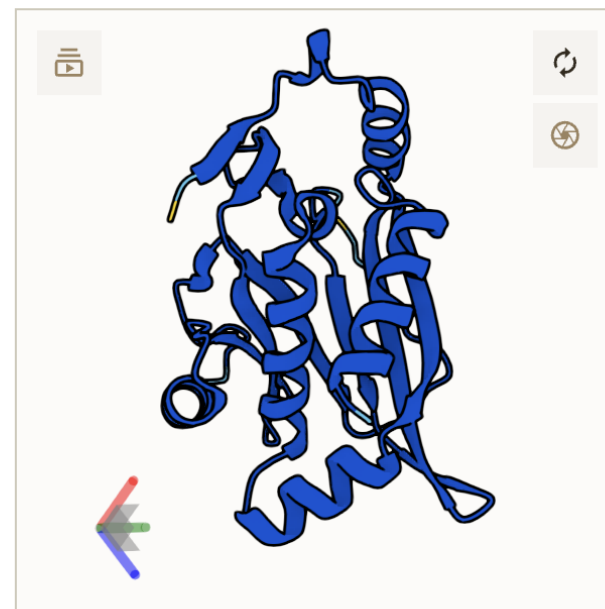
C-type lectin-like domain family
(AF-Q8IZS7-F1)

May function in mediating immune
cell-cell interactions



Apical membrane antigen 1 (AF-
Q3S2X4-F1)

Involved in parasite invasion of
erythrocytes



Chloramphenicol
acetyltransferase (AF-P36883-F1)

Responsible for resistance to the
antibiotic chloramphenicol

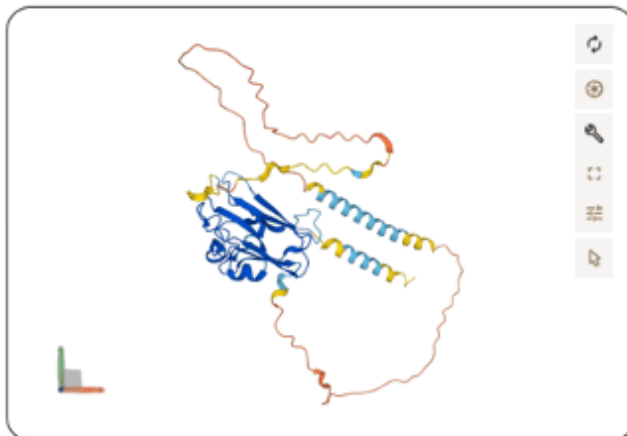
■ Very high (pLDDT > 90) ■ High (90 > pLDDT > 70) ■ Low (70 > pLDDT > 50) ■ Very low (pLDDT < 50)

AF2 confidence metrics, pLDDT

Predicted local distance difference test (**pLDDT**), measures local confidence.

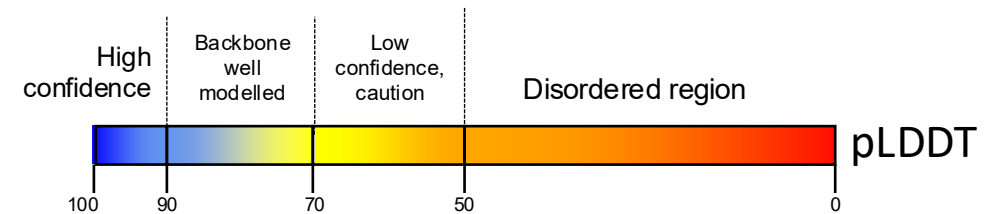
- Scaled from **0 - 100**, with higher scores indicating higher confidence and usually a more accurate prediction.
- It's based on the local distance difference test for C α (IDDT-C α), which assesses the 'correctness of local distances'.

Ephrin-B2



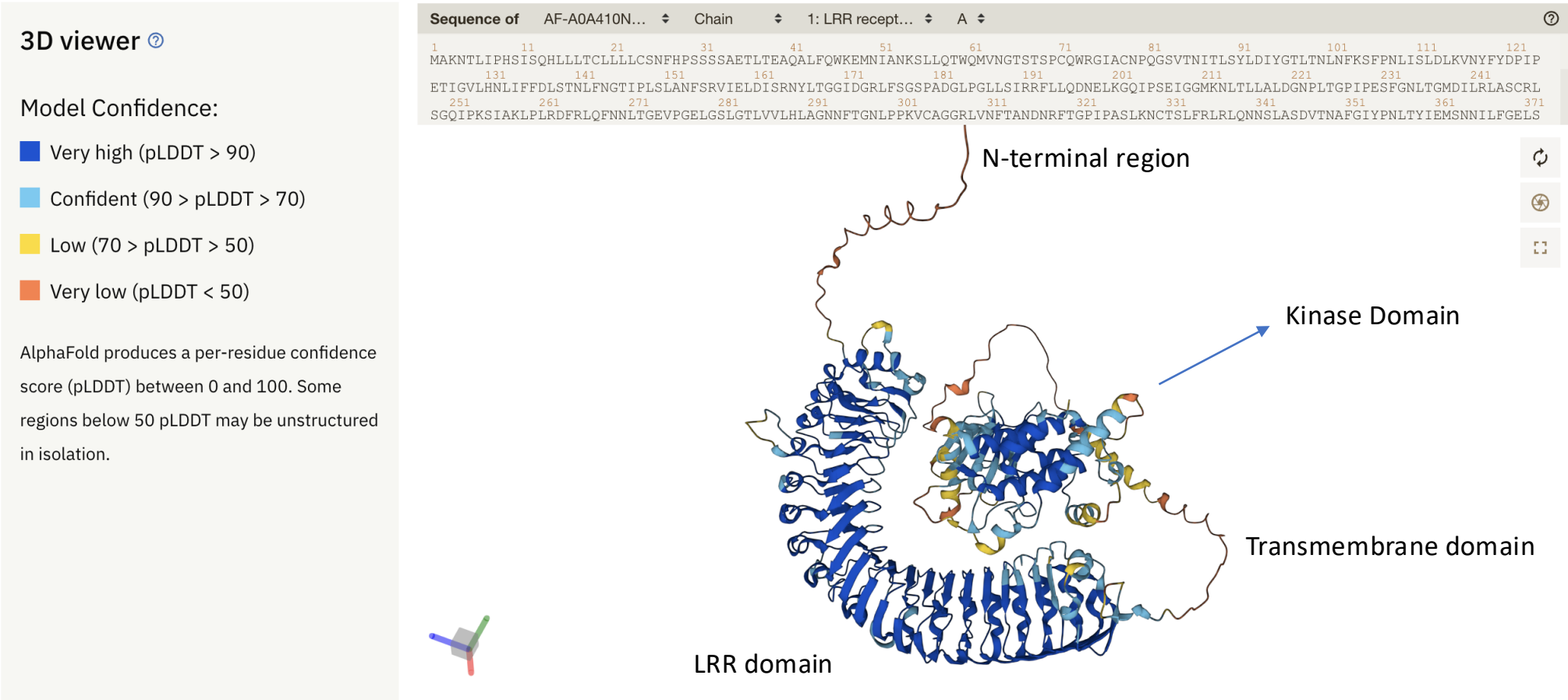
Model Confidence ⓘ

- | | |
|------------------------|-----------------------|
| Very high (pLDDT > 90) | Low (70 > pLDDT > 50) |
| High (90 > pLDDT > 70) | Very low (pLDDT < 50) |



Per-residue confidence

Predicted Local Distance Difference Test (pLDDT)



<https://alphafold.ebi.ac.uk/entry/A0A410N673>

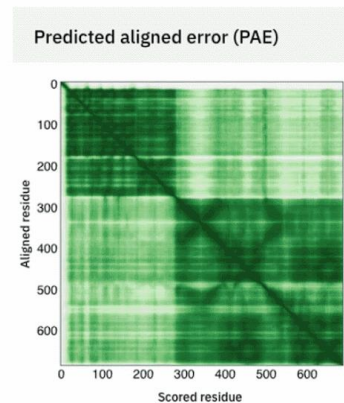
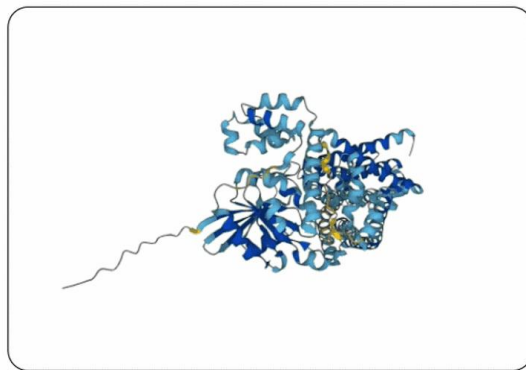
Example: an LRR-RK receptor

AF2 confidence metrics, **PAE**

Predicted alignment error (**PAE**), measures global confidence.

- Measure of the **relative** confidence of the position of **two residues** within the predicted structure.
- PAE is defined as the expected positional error at residue X, in Ångströms (Å), if the predicted and actual structures were aligned on residue Y.

Ferrous iron transport protein B

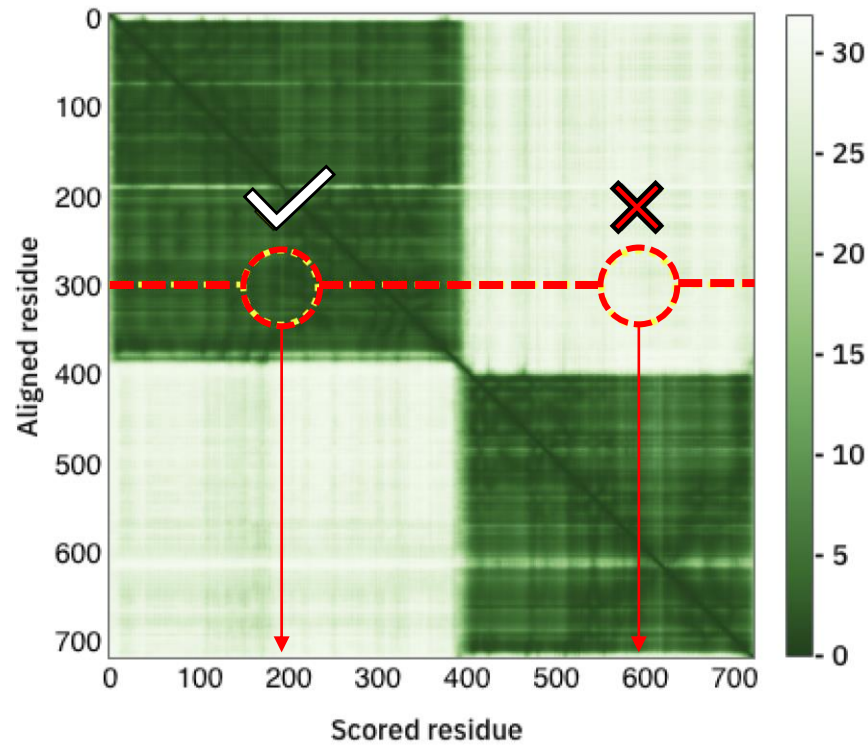


Model Confidence



“PAE is effectively a measure of how confident AlphaFold2 is that **domains are well packed**, and that the **relative placement of the domains in the predicted structure is correct**.”

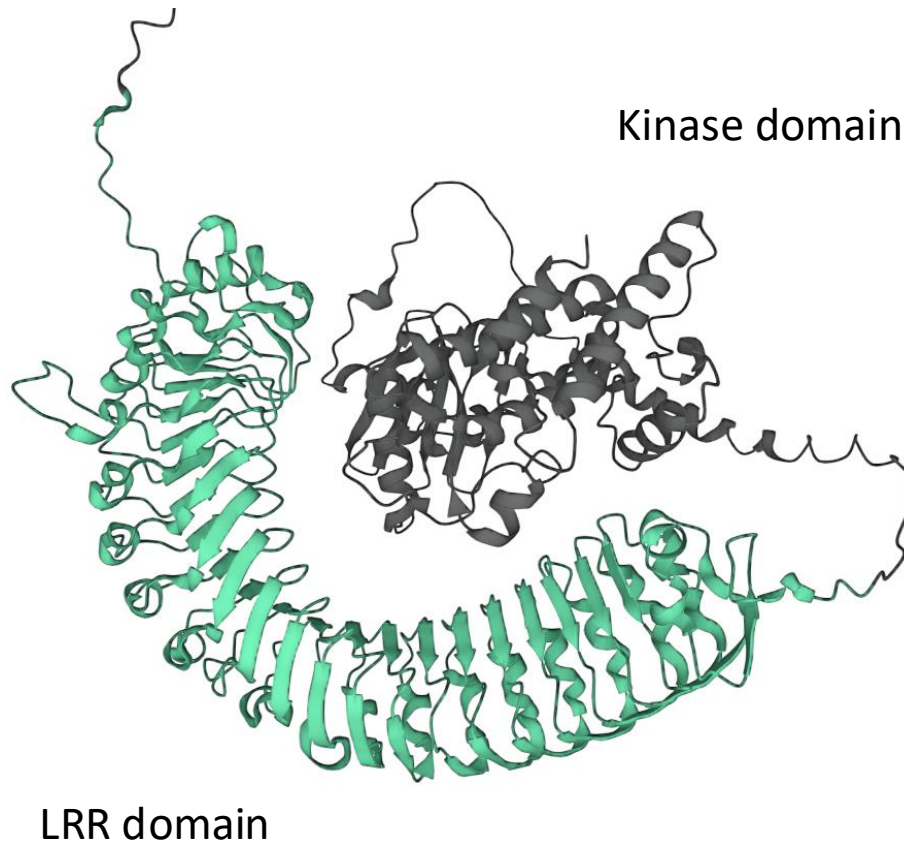
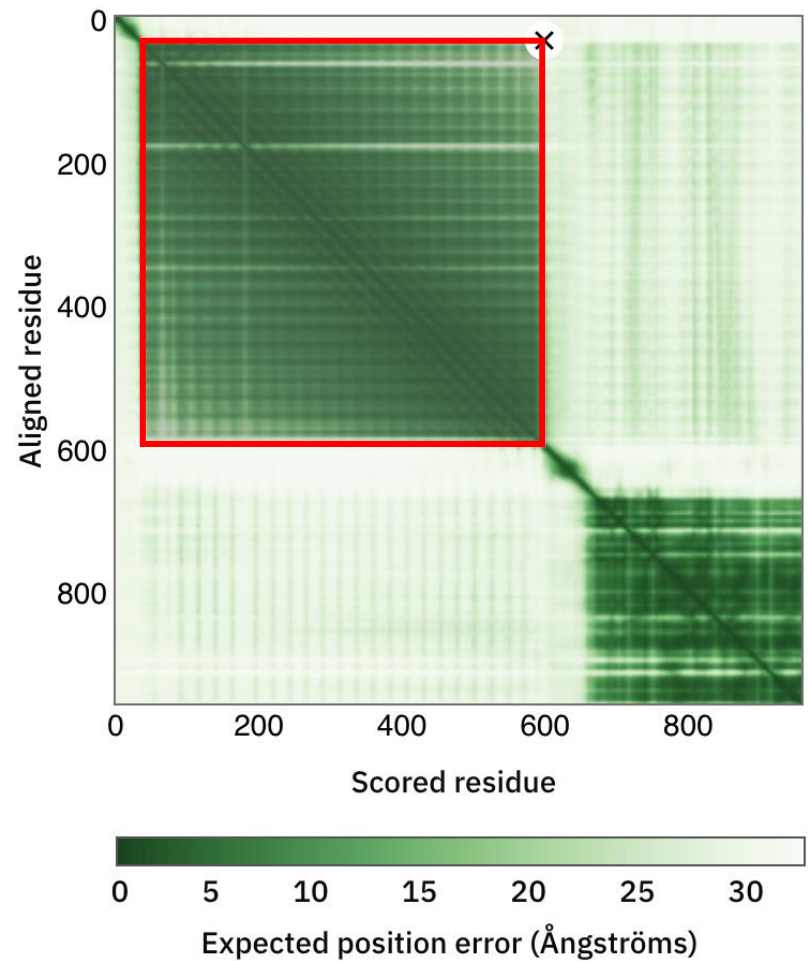
Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)



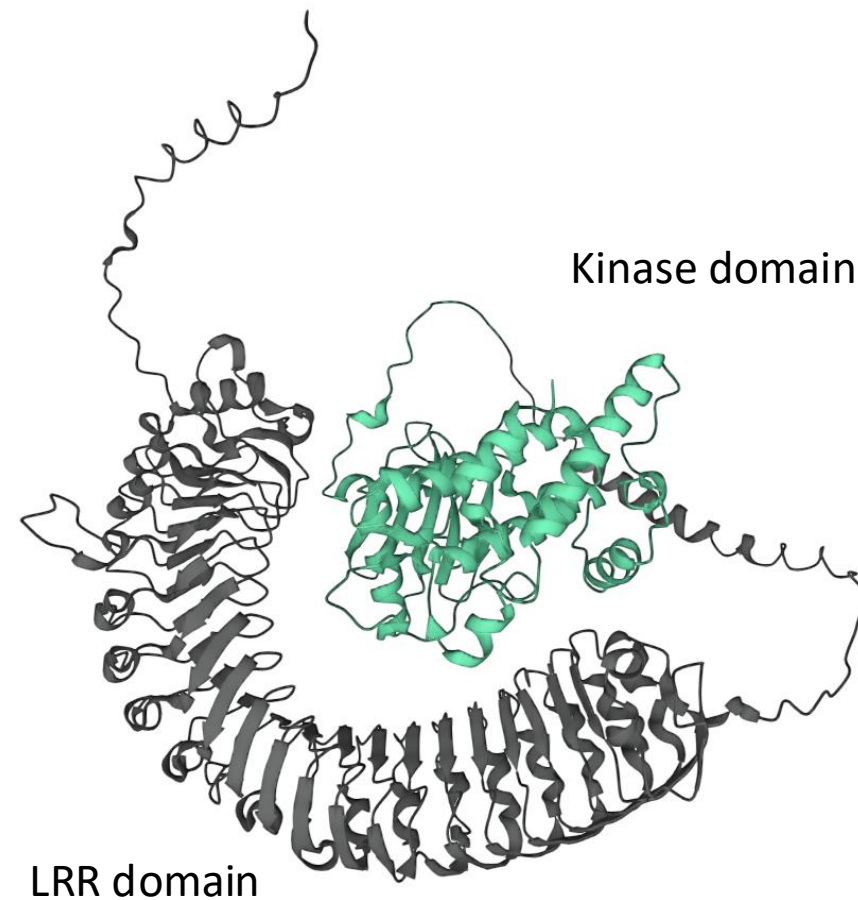
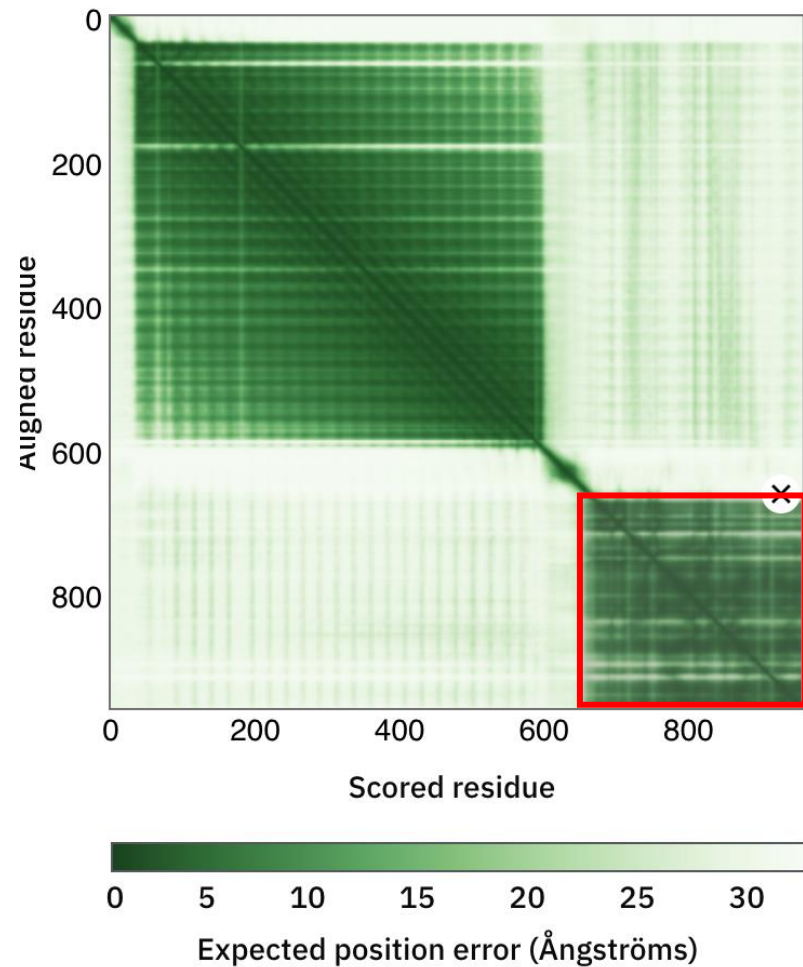
For example, aligning on residue 300:

- We are confident about the relative position of residue 200.
- We are not confident on the relative positioning of residue 600.

Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)

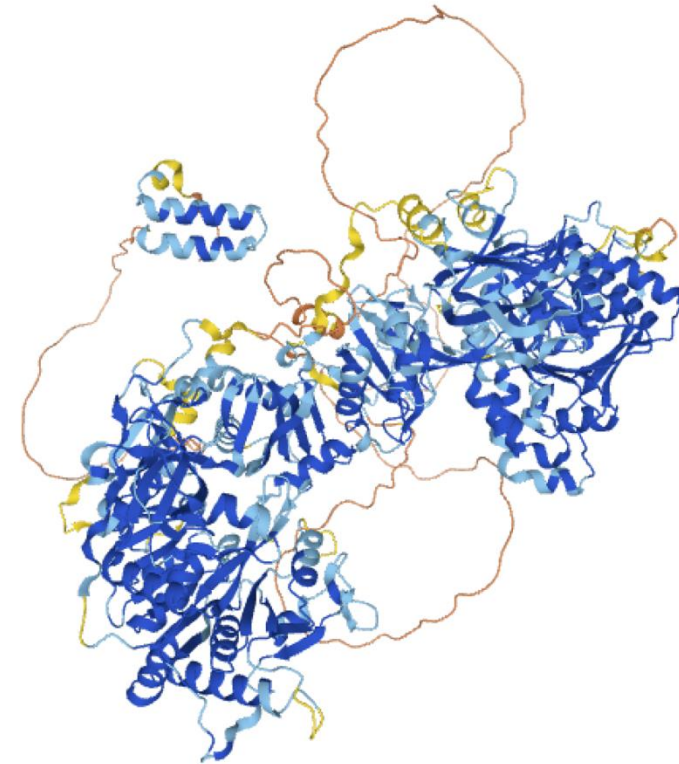
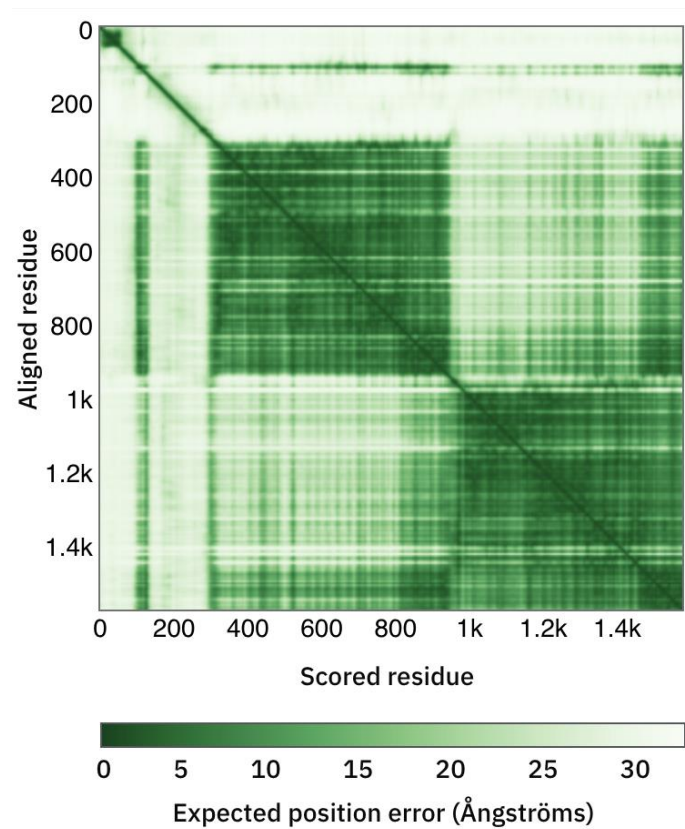


Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)



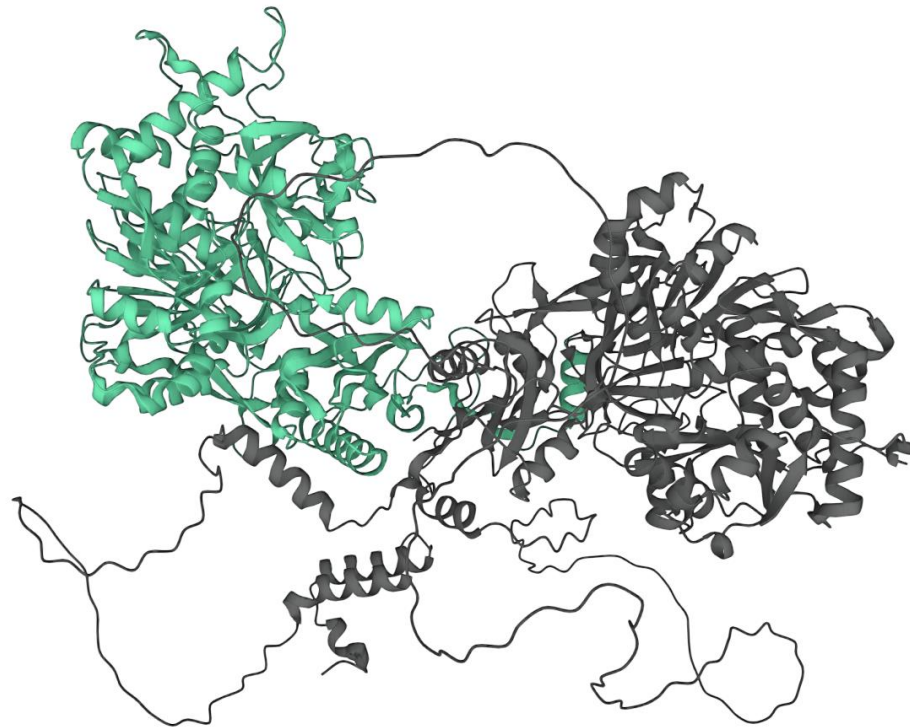
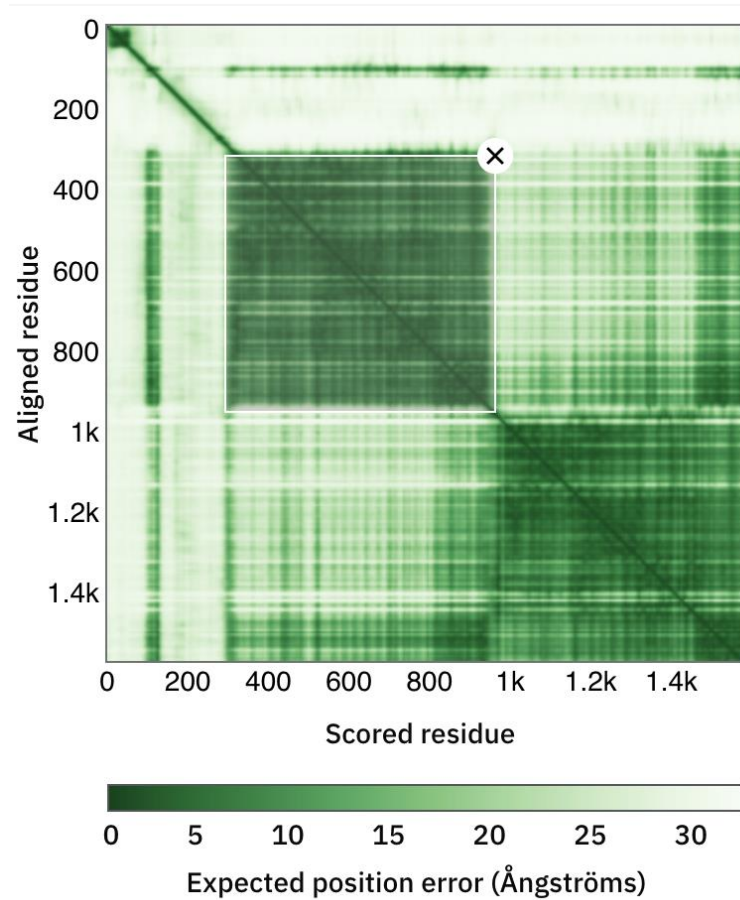
Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)

Another example, human protein *DIP2B* ([Q9P265](#)).

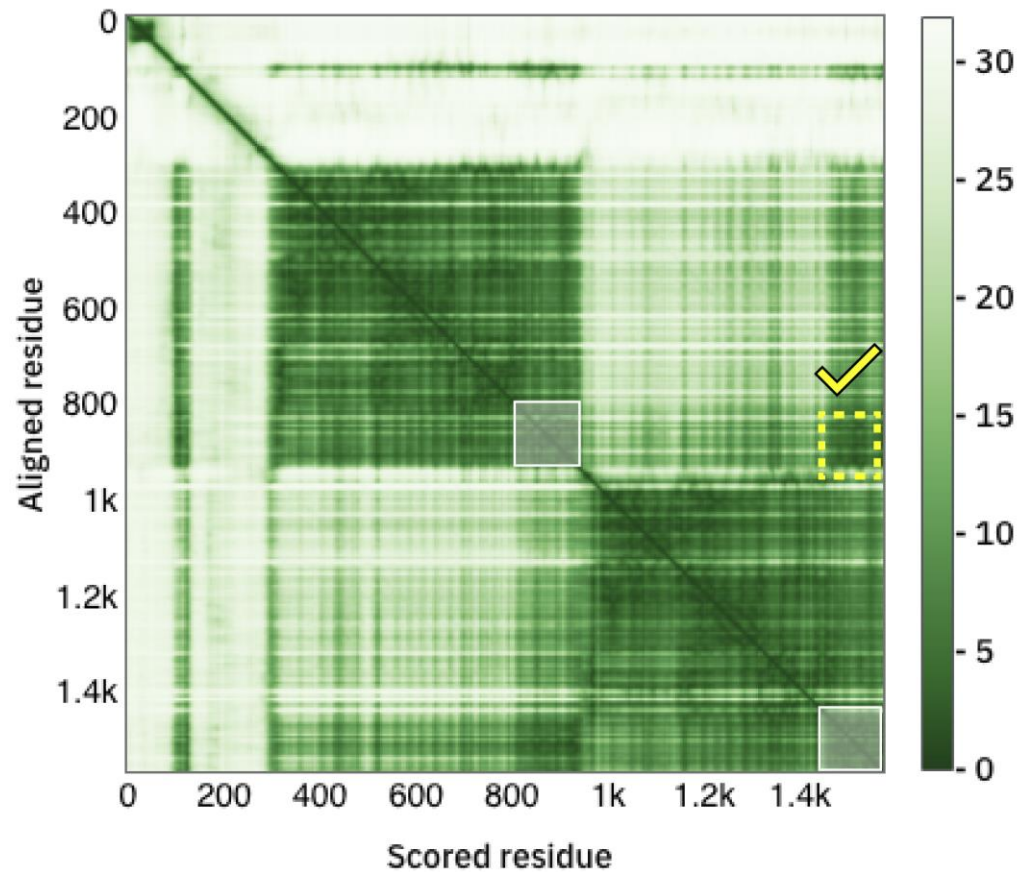


Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)

Another example, human protein *DIP2B* ([Q9P265](#)).

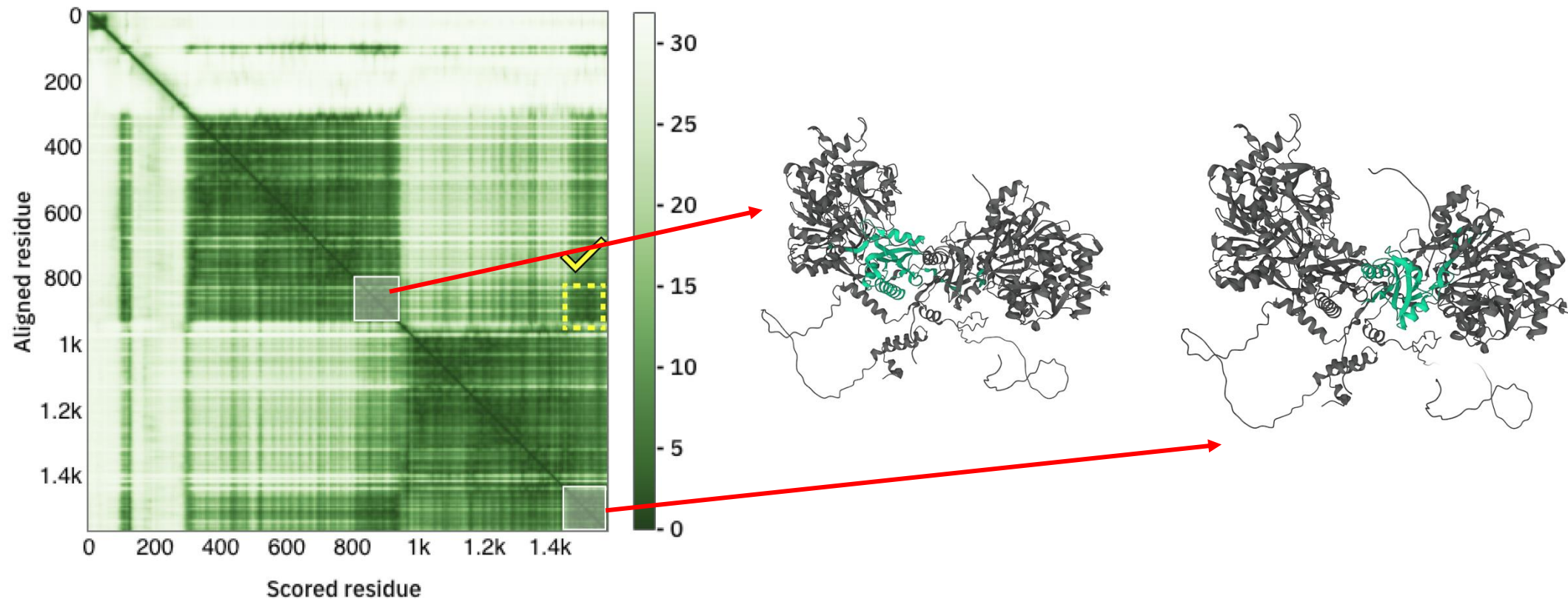


Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)



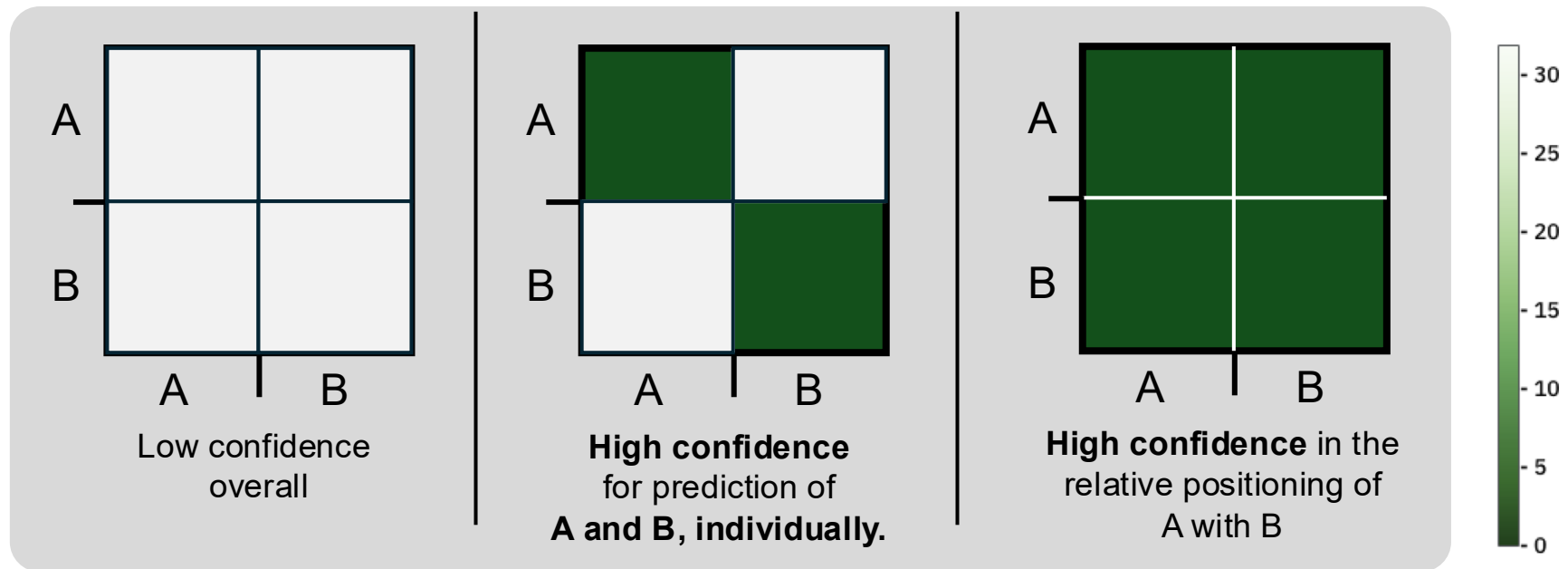
Interpreting the relative position of domains - Predicted Aligned Error plots (PAE plots)

In this example we also see two clear domains, but there is also some confidence in the relative inter-domain position, suggesting packing between small central domains.



Interpreting confidence in relative position of chains

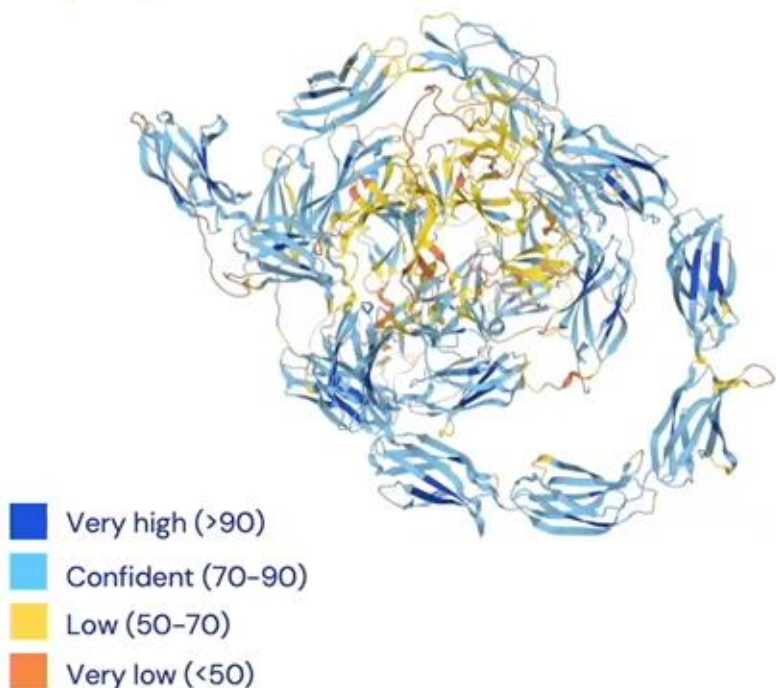
Predicted Aligned Error plots (PAE plots)



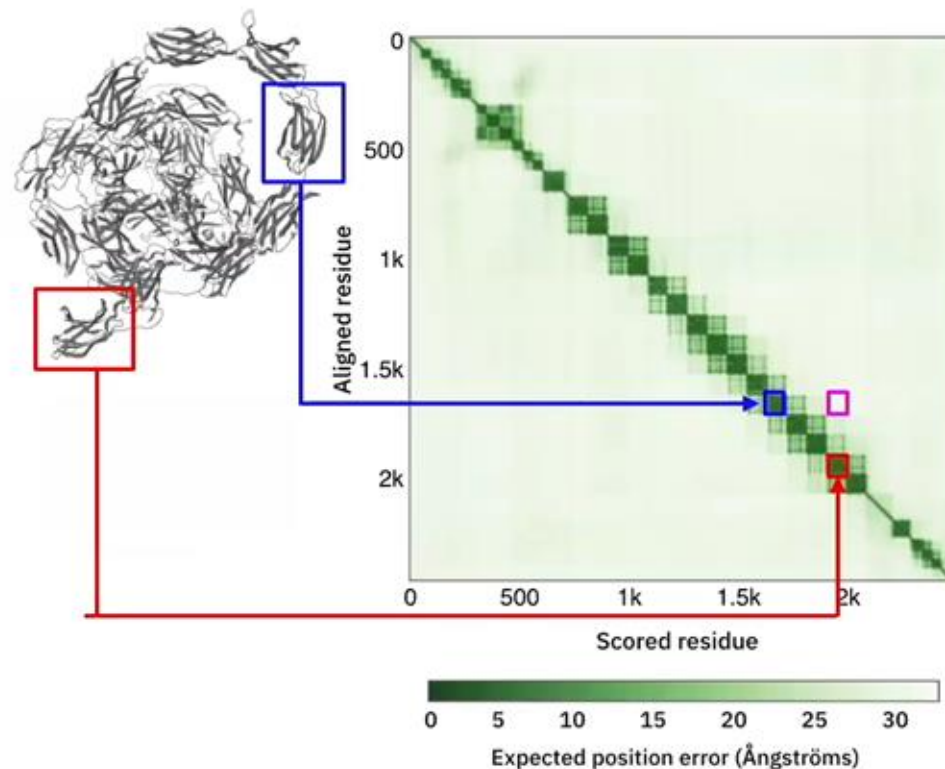
Confidence metrics: pLDDT vs PAE

Training to predict errors

pLDDT: per-residue confidence
(local)



PAE: residue pair confidence (inter-domain/chain)



“As people believe more and more in computational tools and their power, we have a greater and greater responsibility that we report what we know and don’t know in a very cautious and careful way”

Other metrics: ipTM, pTM and model confidence

Monomers

	<u>Range</u>	<u>Interpretation</u>
 • pTM: predicted <u>T</u> emplate <u>M</u> odelling (TM) score.	[0 - 1]	0 < pTM < 0.5 : likely wrong prediction 0.5 > pTM : overall fold might be similar to the true structure

Other metrics: ipTM, pTM and model confidence

		<u>Range</u>	<u>Interpretation</u>
Multimers	Monomers		
	• pTM: predicted <u>T</u>emplate <u>M</u>odelling (TM) score. • ipTM: <u>i</u>nterface <u>p</u>redicted <u>T</u>M-score.	[0 - 1]	0 < ipTM < 0.6 : likely wrong interface 0.6 ≤ ipTM < 0.8 : gray zone 0.8 ≤ ipTM : promising predicted complex

Other metrics: ipTM, pTM and model confidence

	Monomers		Range
Multimers	{	• pTM: predicted <u>T</u> emplate <u>M</u> odelling (TM) score.	
		• ipTM: <u>i</u> nterface <u>p</u> redicted <u>T</u> M-score.	
		• Model confidence = 0.8 (ipTM) + 0.2 (pTM)	[0 – 1]

2.8. Model Confidence

The AlphaFold model provides intrinsic model accuracy estimates in the form of a predicted TM-score (pTM) [1, §1.9.7]. We provide a similar metric for the AlphaFold-Multimer model, but modified to score interactions between residues of different chains, since we are predominantly interested in the accuracy of interfaces. We call this metric Interface pTM, or ipTM. See subsection 7.9 for more details.

In practice we use a weighted combination of pTM and ipTM as our model confidence metric, in order to account for some intra-chain confidence in our model ranking:

$$\text{model confidence} = 0.8 \cdot \text{ipTM} + 0.2 \cdot \text{pTM} \quad (3)$$

Protein complex prediction with AlphaFold-Multimer,
<https://doi.org/10.1101/2021.10.04.463034>

Other metrics: ipTM, pTM and model confidence

		Range	Interpretation
Multimers	Monomers		
	• pTM : predicted <u>T</u> emplate <u>M</u> odelling (TM) score.		
	• ipTM : <u>i</u> nterface <u>p</u> redicted <u>T</u> M-score.		
	• Model confidence = 0.8 (ipTM) + 0.2 (pTM)		
	• ipSAE = <u>i</u> nterface <u>p</u> redicted <u>S</u> tructure <u>A</u> lignment <u>E</u> rror [0 – 1]		 < 0.4 = uncertain 0.4 to 0.6 = possible > 0.6 = likely > 0.8 = highly confident



Other metrics: ipTM, pTM, model confidence, ipSAE

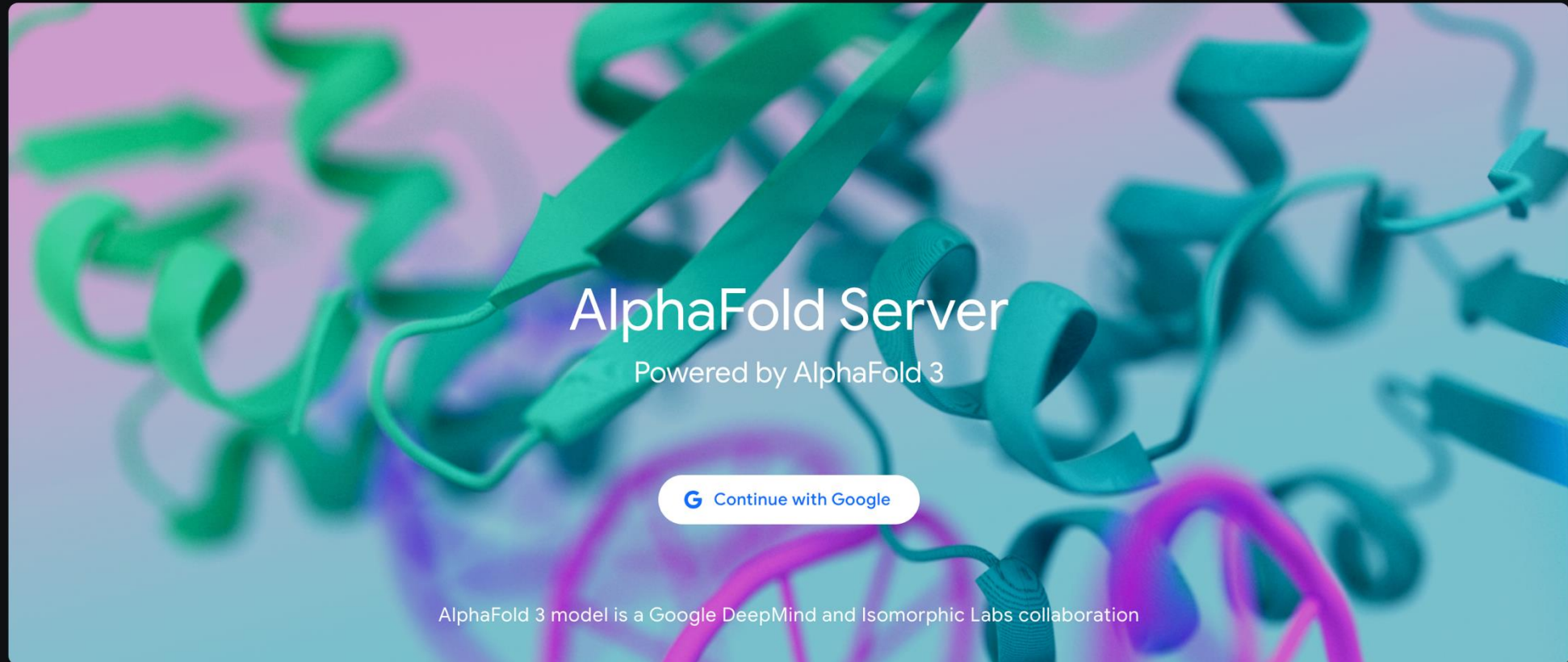
		<u>Range</u>	<u>Interpretation</u>
Multimers	Monomers		
	• pTM : predicted <u>T</u> emplate <u>M</u> odelling (TM) score.		
	• ipTM : <u>i</u> nterface <u>p</u> redicted <u>T</u> M-score.		
	• Model confidence = 0.8 (ipTM) + 0.2 (pTM)		
	• ipSAE = <u>i</u> nterface <u>p</u> redicted <u>S</u> tructure <u>A</u> lignment <u>E</u> rror [0 – 1]		< 0.4 = uncertain 0.4 to 0.6 = possible > 0.6 = likely > 0.8 = highly confident

Note: all these thresholds are ultimately arbitrary...
*Best practice: **have a benchmark dataset with + and – controls.***

AlphaFold-3 release...



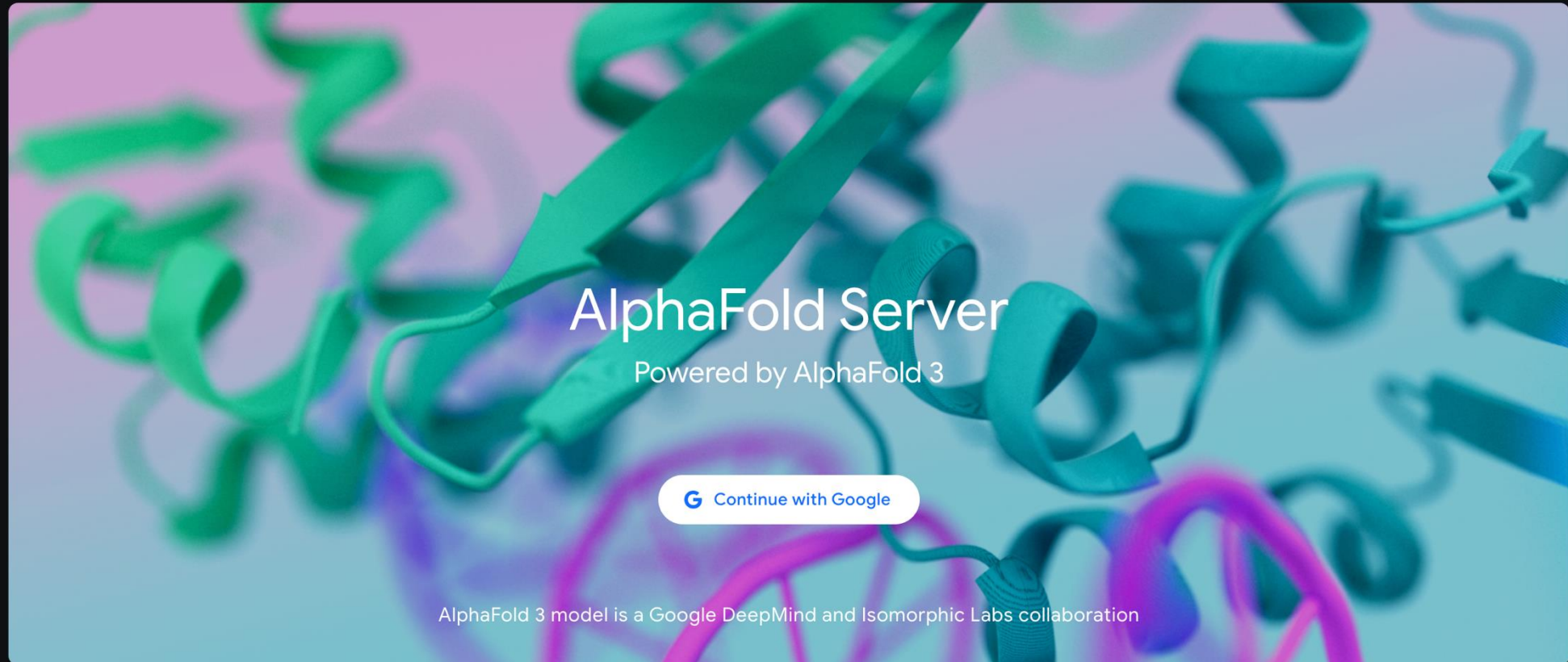
8th of May , 2024



AlphaFold-3 release...



8th of May , 2024



1. Expanded repertoire: Ribonucleic Acids (DNA / RNA), ions, some fatty-acids, co-factors like ATP, NADPH, etc.
2. Allows modelling with post-translational modifications (**PTMs**).
3. Claims improvement over AF-2 for most tasks.

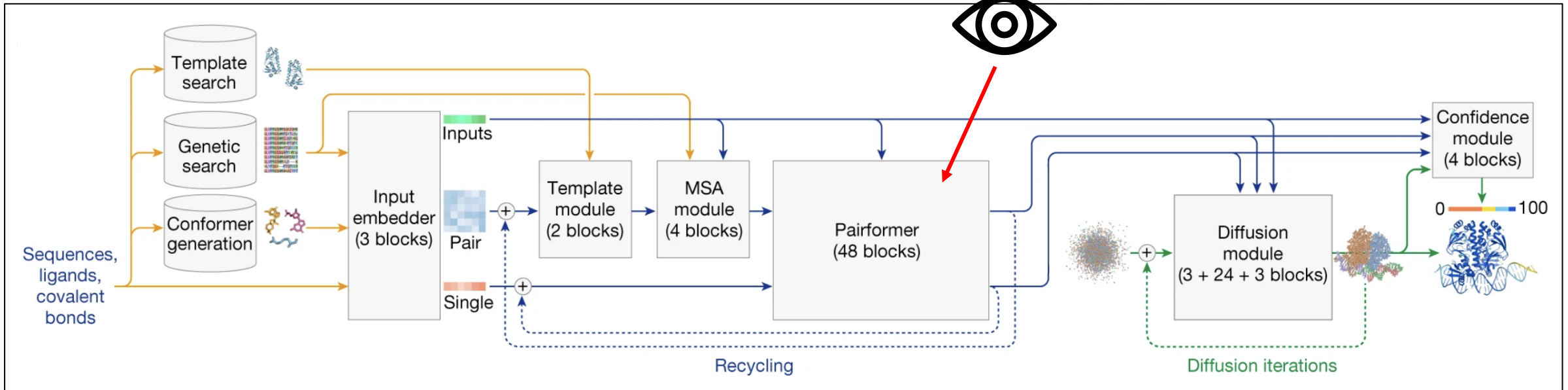
Main differences in architecture

Article | [Open access](#) | Published: 08 May 2024

Accurate structure prediction of biomolecular interactions with AlphaFold 3

[Josh Abramson](#), [Jonas Adler](#), [Jack Dunger](#), [Richard Evans](#), [Tim Green](#), [Alexander Pritzel](#), [Olaf Ronneberger](#), [Lindsay Willmore](#), [Andrew J. Ballard](#), [Joshua Bambrick](#), [Sebastian W. Bodenstein](#), [David A. Evans](#), [Chia-Chun Hung](#), [Michael O'Neill](#), [David Reiman](#), [Kathryn Tunyasuvunakool](#), [Zachary Wu](#), [Akvilė Žemgulytė](#), [Eirini Arvaniti](#), [Charles Beattie](#), [Ottavia Bertolli](#), [Alex Bridgland](#), [Alexey Cherepanov](#), [Miles Congreve](#), ... [John M. Jumper](#)  [+ Show authors](#)

- **CORE MODULE:** AF2 Evoformer replaced with the **Pairformer**, focusing on single and pair representations, **reducing reliance on MSAs**.



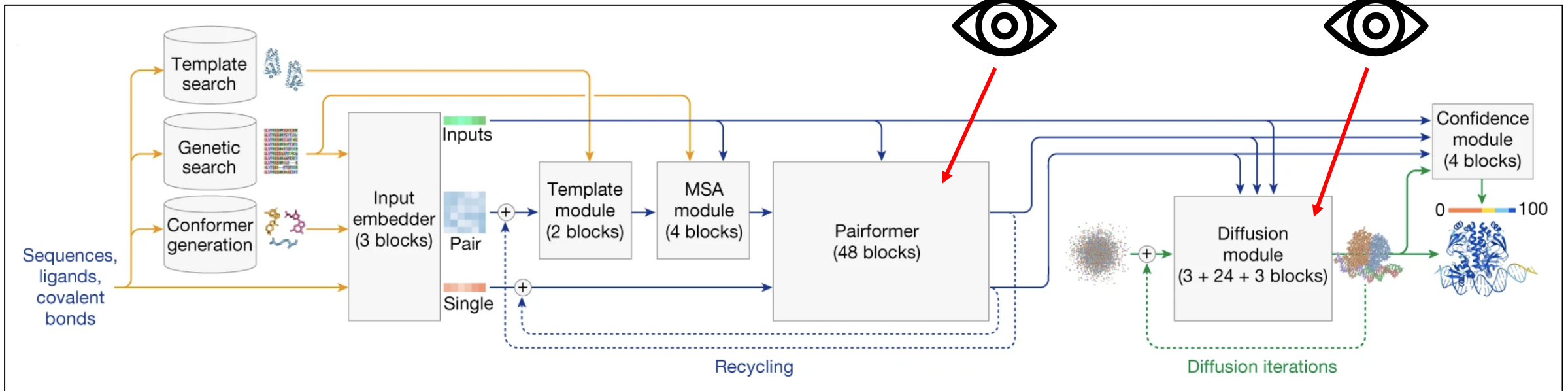
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- **CORE MODULE:** AF2 Evoformer replaced with the **Pairformer**, focusing on single and pair representations, **reducing reliance on MSAs**.
- **STRUCTURE GENERATION:** **Diffusion Module**, a generative model that iteratively refines atomic positions, enabling modelling of complex biomolecular assemblies.



Similarities and differences between Alphafold 2 vs 3

Feature	AlphaFold-2	AlphaFold-3
Release Year	2020	2024
Input Types	Protein sequences	Proteins, DNA, RNA, ligands, ions, and post-translational modifications
Prediction Scope	Single proteins and protein complexes (via AlphaFold-Multimer)	Proteins, protein–protein, protein–nucleic acid, protein–ligand complexes, and covalently modified biomolecules
Architecture	Evoformer + Structure Module	Pairformer + Diffusion-based generative model
MSA Usage	Essential; extensive multiple sequence alignments required	Reduced reliance; capable of handling shallow or no MSAs
Confidence Metrics	pLDDT, PAE, ipTM, pTM (and others: ipSAE, pDockQ, pDockQ2, etc...)	
Open Access	Open-source code and weights available	Limited access via web server; resource-intensive; not fully open-source.
Performance on Complexes	Moderate accuracy for protein–protein interactions	Improved accuracy for protein–protein, protein–nucleic acid, and protein–ligand interactions
Use Cases	Broadly used in academia and industry	Primarily academic research; not licensed for commercial use

Contents

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- 3. Strenghts and Limitations (AF2 and AF3)**

4. Tips & tricks

Uses of AlphaFold (2 & 3)

- Prediction of **new structures of proteins and complexes**.
- **CryoEM** and Crystallography Models. It allows you to start building maps of large complexes. Helps solve structures with **Molecular Replacement** techniques when the data lack sufficient resolution.
- Investigation of **intrinsically disordered proteins** (IDPs). pLDDT allows predicting disordered regions.
- It serves as a guide for **construct design** and protein engineering. Definition of domain boundaries.
- **Oligomerization state prediction**: Dimer? Trimer? Tetramer?, *etc...*
- More...

AF2 and AF3 are routinely used to help solve experimental datasets

The X-ray crystallography phase problem solved thanks to AlphaFold and RoseTTAFold models : a case study report.

Irène Barbarin-Bocahu¹, Marc Graille¹.

¹Laboratoire de Biologie Structurale de la Cellule (BIOC), CNRS, Ecole polytechnique, Institut Polytechnique de Paris, F-91128 Palaiseau, France

Keywords

Structural biology / Phase problem / Molecular replacement / Machine learning 3D models / AlphaFold

Accelerating crystal structure determination with iterative AlphaFold prediction

Thomas C. Terwilliger^{1,2}, Pavel V. Afonine³, Dorothee Liebschner³, Tristan I. Croll⁴, Airlie J. McCoy⁴, Robert D. Oeffner⁴, Christopher J. Williams⁵, Billy K. Poon³, Jane S. Richardson⁵, Randy J. Read⁴ and Paul D. Adams^{3,6}

¹New Mexico Consortium, Los Alamos, NM 87544, USA

²Los Alamos National Laboratory, Los Alamos, NM 87545, USA

³Molecular Biophysics & Integrated Bioimaging Division, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA.

⁴Department of Haematology, Cambridge Institute for Medical Research, University of Cambridge, Hills Road, Cambridge CB2 0XY, United Kingdom

⁵Department of Biochemistry, Duke University, Durham, North Carolina, 27710

⁶Department of Bioengineering, University of California, Berkeley, Berkeley, CA 94720, USA

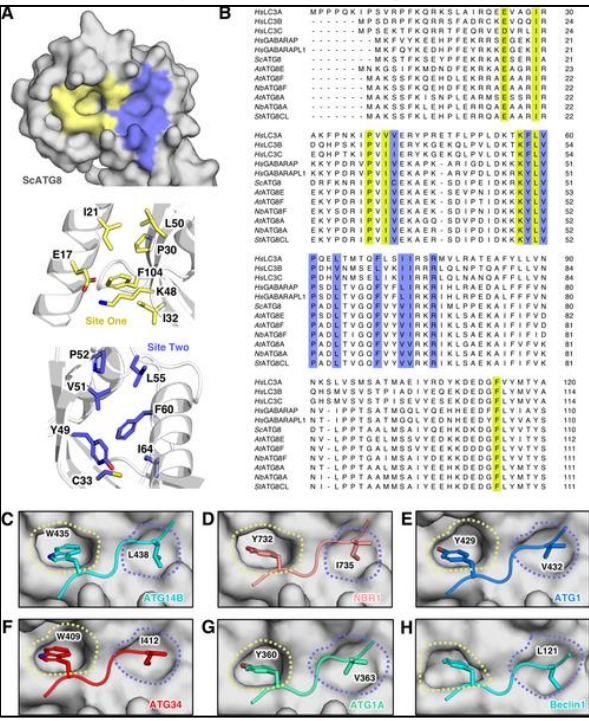
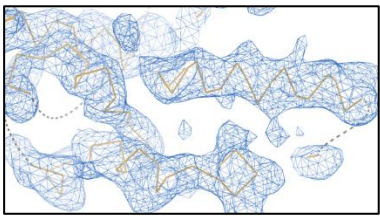
Experimental structure determination can be accelerated with AI-based structure prediction methods such as AlphaFold. Here we present an automatic procedure requiring only sequence information and crystallographic data that uses AlphaFold predictions to produce an electron density map and a structural model. Iterating through cycles of structure prediction is a key element of our procedure: a predicted model rebuilt in one cycle is used as a template for prediction in the next cycle. We applied this procedure to X-ray data for 215 structures released by the Protein Data Bank in a recent 6-month period. In 87% of cases our procedure yielded a model with at least 50% of C α atoms matching those in the deposited models within 2Å. Predictions from our iterative template-guided prediction procedure were more accurate than those obtained without templates. We suggest a general strategy for macromolecular structure determination that includes AI-based prediction both as a starting point and as a method of model optimization.

A non-exhaustive list of AF2 use-cases in the plant immunity field...

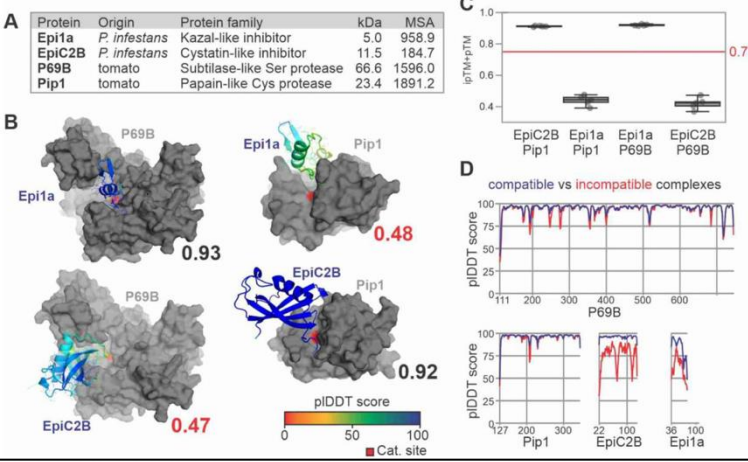
AlphaFold2-multimer guided high-accuracy prediction of typical and atypical ATG8-binding motifs

Tarhan Ibrahim, Virendrasinh Khandare, Federico Gabriel Mirkin, Yasin Tumtas, Doryen Bubeck, Tolga O. Bozkurt

Published: February 8, 2023 • <https://doi.org/10.1371/journal.pbio.3001962>

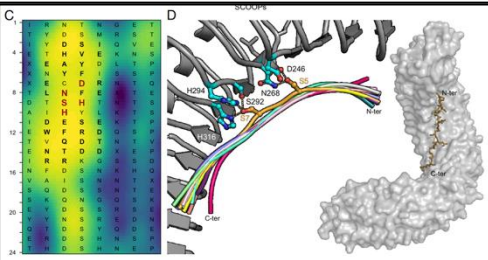


FIGURES



Leveraging coevolutionary insights and AI-based structural modeling to unravel receptor-peptide ligand-binding mechanisms

Simon Snoeck^{a,1}, Hyun Kyung Lee^b, Marc W. Schmid^c, Kyle W. Bender^a, Matthias J. Neeracher^a, Alvaro D. Fernández-Fernández^a, Julia Santiago^b, and Cyril Zipfel^{a,d,1}



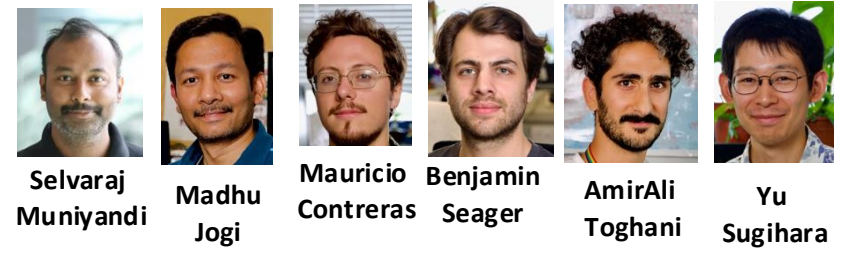
Cyril Zipfel

Alphafold-multimer predicts cross-kingdom interactions at the plant-pathogen interface

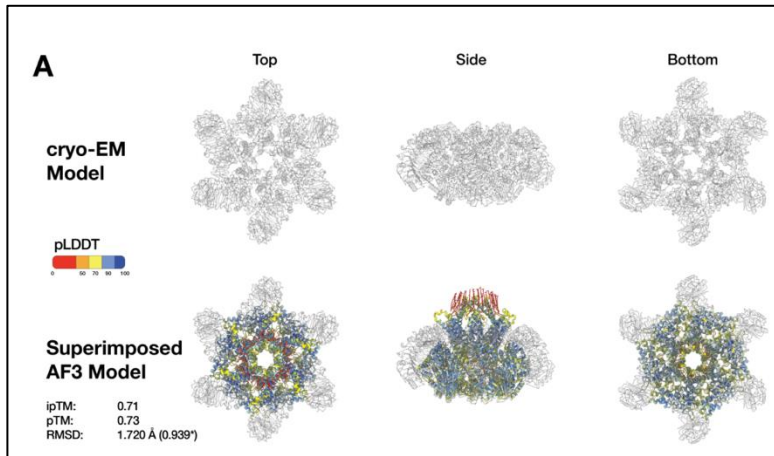
Felix Homma¹, Jie Huang¹, Renier A. L. van der Hoorn^{*}

In the Kamoun group and at TSL...

- Molecular replacement with an AF2 model to help map the structure of the nematode effector SPRYSEC15.
- Driving so many projects by guiding our hypotheses, which we follow up by mutagenesis and functional assays.



'Resistosome' modelling and oligomeric state predictions...



- Impressively high **coherence** between prediction and Cryo-EM-resolved complex.
- **Oligomeric state** predictions?
- An emerging new tool **to rank and prioritize** which immune receptors may be forming very different complexes from the ones already described.

Read more... **PMID:** 39504373



Madhu
Jogi



AmirAli
Toghani



Mauricio
Contreras



Andres
Posbeyikian

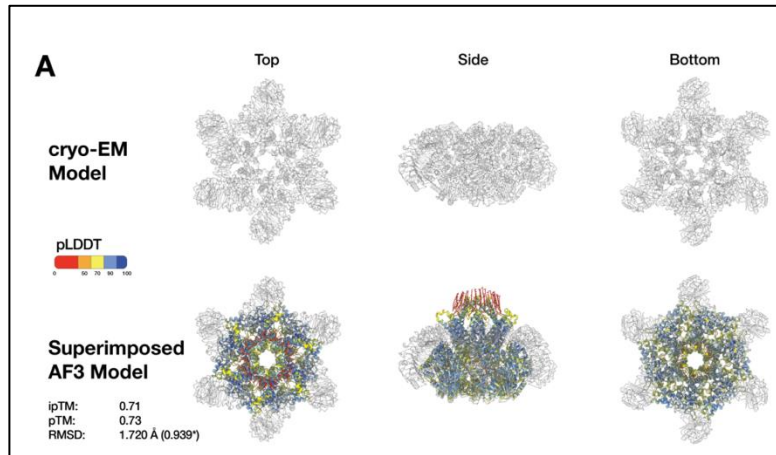


Michael
Webster

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et. al.

'Resistosome' modelling and oligomeric state predictions...

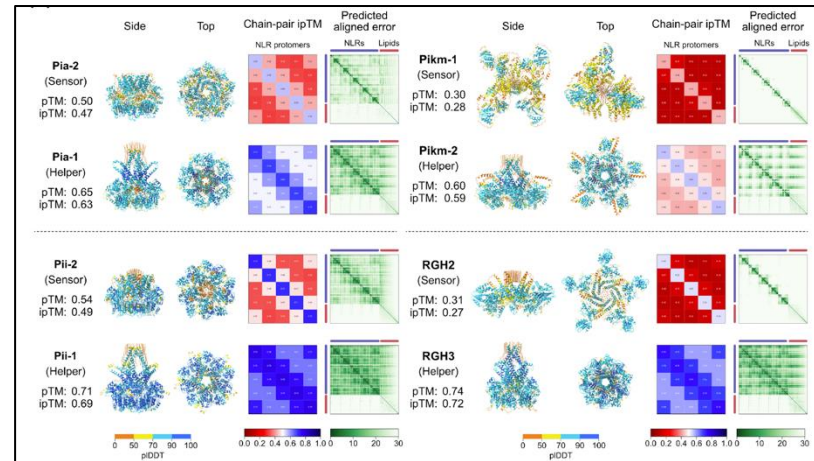


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Read more... **PMID: 39504373**

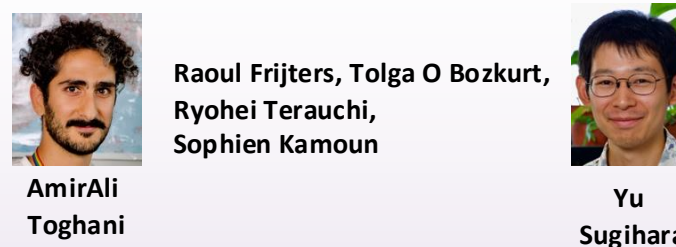


Functional prediction and classification...

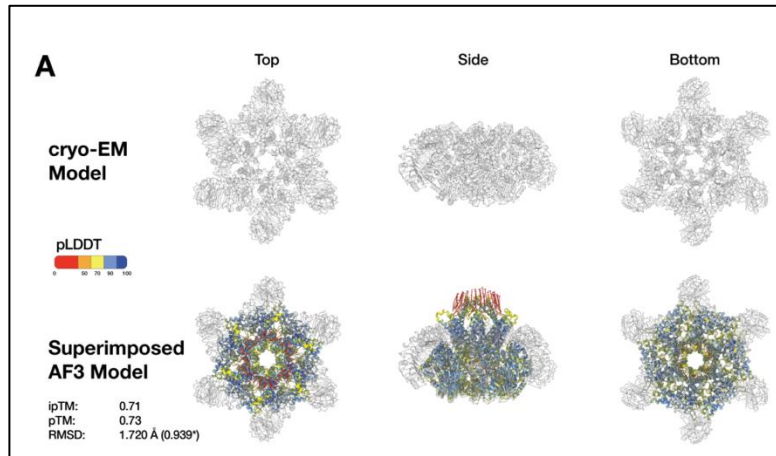


- Prediction of **funnel-like** structures.
- Functionally classify immune receptors into the type that can likely oligomerize into 'resistosomes' (and execute an immune response) and the ones that likely cannot.

Read more... **DOI: 10.1101/2024.11.24.625045**



'Resistosome' modelling and oligomeric state predictions...

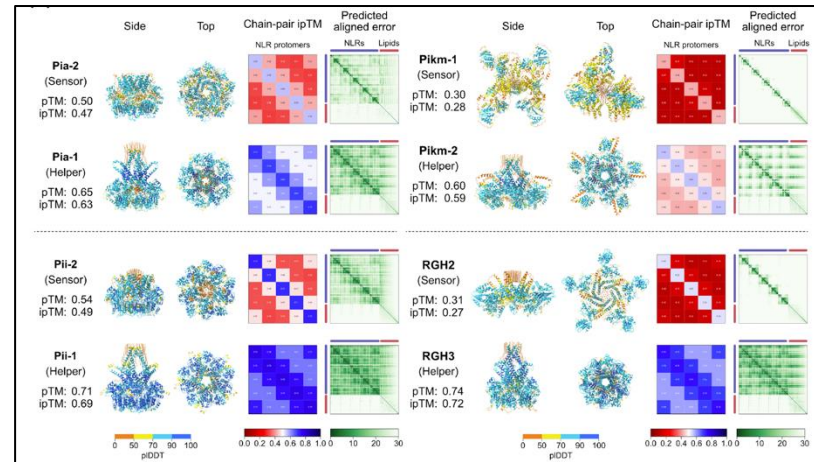


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Read more... **PMID: 39504373**

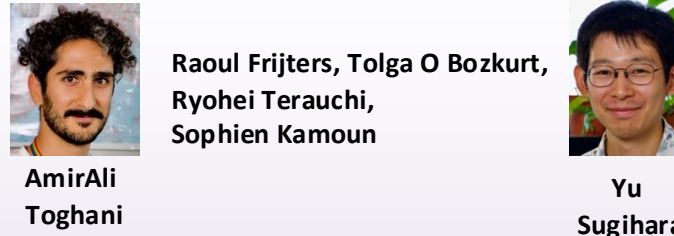


Functional prediction and classification...

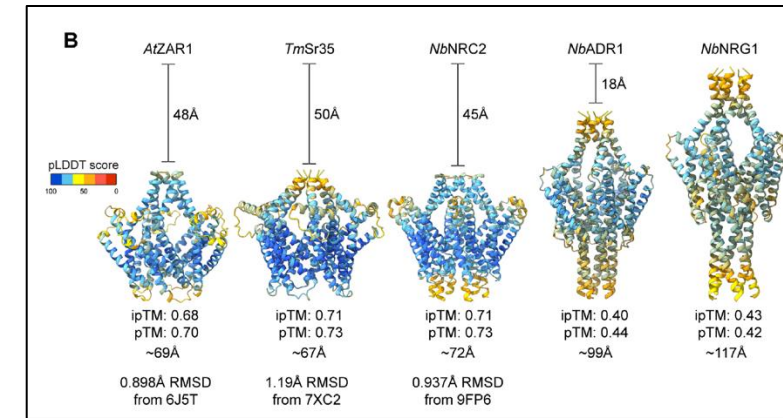


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Read more... **DOI: 10.1101/2024.11.24.625045**



Hypothesis driver / filling gaps in our understanding



Collaboration with **Bozkurt Lab** @ Imperial College London

- AF3 models are **supporting** observations that some activated immune receptors are deployed at double-membraned organelles.
- An extended funnel may be necessary to penetrate the two membranes. This drove the hypothesis which was quickly validated through domain-swapping experiments!

Read more... **DOI: 10.1101/2024.09.19.613839**

Tarhan Ibrahim Enoch Yuen Hung-Yu Wang Freddie King AmirAli Toghani et. al...

Some limitations...

- All models **must** subsequently be **tested** with **functional PPI assays**. AF only provides models, which in many cases enable making hypotheses. A high confidence multimer complex does not imply that there is **high affinity** between the modeled proteins. To study the affinity, empirical protein-protein binding assays need to be carried out.
- It is not useful for modeling **highly variable proteins** such as antibodies. There are tools that are appearing for this (*i.e.* NanoNet 2021, ABlooper, etc...).
- It can be used to predict **intrinsically disordered** regions/proteins, which are typically modeled as ribbons, but without confidence or consistency in the predicted structure.
- Not good at modeling fold-switching proteins. When a protein is known to have **multiple conformations**, AF usually only produces one of them. The output conformation cannot be controlled reliably. New tools are coming out that address this.
- AF was not validated to predict the effect of **mutations**. In particular, AF is not expected to produce an unfolded protein structure given a sequence containing a **destabilizing point mutation**.
- **AF2** does not predict positions of any **non-protein binding partners** found in experimental structures (cofactors, metals, ligands, ions, DNA/RNA or post-translational modifications). However, **AF3** does support many ligands and co-factors, as well as post-translational modifications.

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Tips and tricks for modelling more efficiently

AF-2

- If the region of interest is a **specific domain** of a multi-domain protein, it can be **modeled in isolation** and this will save time and compute units.
- When modeling in an exploratory manner, it is advisable to do so with **3 rounds of recycling**, and only with 12 recycles in final modelling instances.

AF-3

- When modeling membrane-bound proteins, **oleic acids** may be used as stand-in for phospholipids.
- On web-server, cut-off dates for structural template searches in the PDB → This may help prevent a bias caused by a deposited structure.

AF-2 and 3

- Provide an input sequence that reflects your mature protein. *i.e*: remove any signal peptides and other cleaved regions present in the immature polypeptide (for secreted and subsequently cleaved proteins).
- If the protein to be modeled has an intrinsically disordered region, the sequence can be trimmed to facilitate modeling, and this will generally not affect modeling of the rest of the chain.

Thank you for listening!

Q&A

Any additional questions:
Andres.Posbeyikian@tsl.ac.uk



Programme

The same two-day programme runs at both venues (Espace Zmorda, then ISBS).

Day 1 Predict & Interpret

08:30–09:30	Registration
09:30–09:45	Welcome & Introductions
09:45–10:15	Introduction to AI for Genomics
10:15–11:00	Theoretical Introduction to AlphaFold
11:00–11:20	Break
11:20–12:30	 Practical 1: Predict
12:30–13:30	Lunch
13:30–15:00	<i>GOHREP it!</i> Interactive discussions
15:00–15:15	Break
15:15–17:00	 Practical 2: Interpret
17:00–17:15	Q&A

Day 2 Compare & Design

09:00–09:30	Day 2 Welcome & Registration
09:30–11:15	 Practical 3: Compare
11:15–11:30	Break
11:30–12:30	Model your own proteins
12:30–13:30	Lunch
13:30–14:45	 Practical 4: Design
14:45–15:05	Break
15:05–16:15	Genomics in an AI world
16:15–16:30	Summary
16:30–17:00	Q&A
17:00	Meeting close